

**HIGH PRIORITY****CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A**

Owner: County of Hawaii
Contractor: Nan, Inc.
Project Name: Hilo WWTP Phase 1
Submittal Title: 600V or Less Wire and Cable
TO: COH
From: Nan Inc.

Project No.: WW-4705R
Submittal Number: 16123

☐ FOR INFORMATION ONLY

Specification No. and Subject of Submittal / Equipment Supplier			
Spec:	16123	Paragraph:	1.03 B
Authorized By:	Mass Electric	Date Submitted:	10/1/2025

Submittal Certification				
Check Either (A) or (B):				
☒ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.			
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.			
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.				
General Contractor's Reviewer's Signature: 				
Printed Name and Title: Mark McCarthy QC				
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.				
Firm:	Signature:	Date Returned:		

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.DATE RECEIVED 10/1/2025
SPECIFICATION SECTION # 16123
SPECIFICATION 600V or less wire and cable
PARAGRAPH 1.03 B
DRAWING n/a
SUBCONTRACTOR Mass Electric
SUPPLIER n/a
MANUFACTURER n/aCERTIFIED BY CQCM or Designee: 

Submittal



SUB Reference No : MECI-SUB-0011

Status: GC AND CLIENT REVIEW

For Action: Darrin Lee, MECI
David Wieseler, MECI

Project: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT

Subject: 16123 - 600-VOLT OR LESS WIRES AND CABLES

Submittal for 600-volt class wire and cable, 600-volt VFD cable, instrumentation class wire and cable, network cable, cable ties, and lubricant.

Discipline: ELEC **Area:** 00

Submitted For: Approval

Specification Reference : 16123 - 600-VOLT OR LESS WIRES AND CABLES

Paragraph No.: 1.03.B

Description : Submittal for 600-volt class wire and cable, 600-volt VFD cable, instrumentation class wire and cable, network cable, cable ties, and lubricant.

Discipline: ELEC **Area:** 00

SUB by:	Hannah Anderson, MECI	On: 26 September 2025
Created By:	Hannah Anderson, MECI	On: 26 September 2025, 01:49:02 PM -10:00
Last Edited By:	Darrin Lee, MECI	On: 29 September 2025, 12:41:02 PM -10:00

01 - QC/Project Engineer Review

Approved without comment.

QC/Project Engineer Review by:	David Wieseler, MECI	On: 26 September 2025
Created By:	David Wieseler, MECI	On: 26 September 2025, 03:20:15 PM -10:00
Last Edited By:	David Wieseler, MECI	On: 26 September 2025, 03:20:15 PM -10:00

02 - Project Manager Review

updated submittal package attached. Removed previous submittal package.

Attachments:	HWWTP - 16123 - 600-VOLT OR LESS WIRE AND CABLE.pdf	
Project Manager	Darrin Lee, MECI	On: 26 September 2025
Review by:		
Created By:	Darrin Lee, MECI	On: 26 September 2025, 10:00:28 PM -10:00
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MASS ELECTRIC CONSTRUCTION COMPANY PRODUCT DATA SUBMITTAL

REVISION	DATE	PREPARED BY	APPROVED BY
0	9/26/2025	Hannah Anderson	David Wieseler

HILO WASTEWATER TREATMENT PLANT REHABILITATION AND REPLACEMENT PROJECT



SPECIFICATION - 16123 600-VOLT OR LESS WIRES AND CABLES

DOCUMENT REVISION LOG						
Revision Number	Revision Date	Description	Approvals			
			PE INITIAL		PM INITIAL	
0	9/26/2025	Issued for Review	DCW	9/26/2025		

COMMENTS AND CLARIFICATIONS

1. Instrumentation Class Cable (Spec 2.05.D)

We are submitting Type TC-ER cables where available, in lieu of Type TC cables as specified in 16123-2.05.D.1. ER-rated TC cables provide enhanced crush and impact ratings, and flexibility in allowable exposure, which will support constructability.

2. Instrumentation Class Cable (Spec 2.05.D)

Per specification, instrumentation class cables are required to have a temperature rating of 90°C in dry locations and 75°C in wet locations. The submitted cable is rated for 90°C in both dry and wet locations, exceeding the wet location requirement. This enhanced rating is provided at no cost to the County.

3. Network Cables (Spec 2.0.5.E.1)

Submittal for 600V-rated CAT6 (copper Ethernet) network cables will be provided at a later date, contingent upon the final configuration of the motor control centers. These cables may be required based on system design. This submittal includes 300V-rated CAT6 cable.

4. RS-485 Cable (Spec 2.05.E.3)

The nominal outer diameter of the RS-485 cable jacket does not meet the specified value. We request an exception to allow for the larger OD of a 4-wire shielded twisted pair RS-485 cable.

5. RS-485 Cable (Spec 2.05.E.3)

The submitted RS-485 cable complies with the bend radius requirement of at least 10 times the cable diameter. With an OD of 0.350 inches, the minimum bend radius is 3.5 inches, which is met by the proposed cable.

6. Wire Ties (Spec 2.06.A)

Due to product availability and frequency of use, we are submitting both T&B Ty-Rap and Panduit cable ties. Both products meet the requirements for UV resistance, weather durability, and expected service life.

7. Wire Markers (Spec 2.06.B)

Wire markers will be included in the submittal for Section 16075-Identification for Electrical Systems.

8. Wire-pulling Lubricant (Spec 3.01.B.1.b)

We are submitting both Polywater J and Polywater LZ wire-pulling lubricants due to product availability and frequency of use. Both products meet the required performance standards.

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9/26/2025

SECTION 16123

600-VOLT OR LESS WIRES AND CABLES

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes:
 - 1. 600-volt class or less wire and cable.

1.02 REFERENCES

- A. Abbreviations:
 - 1. AWG: American wire gauge.
 - 2. BCCS: Bare copper-covered steel.
 - 3. CPE: Chlorinated polyethylene.
 - 4. FEP: Fluorinated ethylene propylene.
 - 5. FHDPE: Foam high-density polyethylene.
 - 6. FPE: Foam polyethylene.
 - 7. OD: Outside diameter.
 - 8. PVC: Polyvinyl chloride.
 - 9. XHHW: Cross-linked high heat water resistant insulated wire.
- B. Definitions:
 - 1. Definitions of terms and other electrical considerations as set forth in the:
 - a. ASTM.
 - b. ICEA.
- C. Standards:
 - 1. ASTM International (ASTM):
 - a. B3 - Standard Specification for Soft or Annealed Copper Wire.
 - b. B8 - Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft.
 - 2. CSA International (CSA).
 - 3. Insulated Cable Engineer's Association (ICEA):
 - a. S-90-661 - Individually Unshielded Twisted Pair Indoor Cables for Use in Communication Wiring Systems.
 - 4. National Electrical Code (NEC).
 - 5. National Electrical Manufacturers Association /Insulated Cable Engineers Association (NEMA/ICEA):
 - a. NEMA WC 57/ICEA S-73-532 - Standard for Control, Thermocouple Extension, and Instrumentation Cables.
 - b. NEMA WC 66/ICEA S-116-732 - Standard for Category 6 and 6A, 100 Ohm, Individually Unshielded Twisted Pairs, Indoor Cables (With or Without an Overall Shield) for Use in LAN Communication Wiring Systems.
 - c. NEMA WC 70/ICEA S-95-658-1999 - Standard for Power Cables Rated 2000 Volts or Less for the Distribution of Electrical Energy.

6. National Fire Protection Association (NFPA):
 - a. 262 - Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air-Handling Spaces.
7. Telecommunications Industry Association (TIA):
 - a. 568.2-D - Balanced Twisted-Pair Telecommunications Cabling and Components Standard.
 - b. 569-B - Commercial Building Standards for Telecommunications Pathways and Spaces.
8. Underwriter's Laboratories Inc., (UL):
 - a. 44 - Standard for Thermoset-Insulated Wires and Cables.
 - b. 1569 - Standard for Metal-Clad Cables.
 - c. 1666 - Standard for Test for Flame Propagation Height of Electrical and Optical-Fiber Cables Installed Vertically in Shafts.
 - d. 2225 - Cables and Cable-Fittings for Use in Hazardous (Classified) Locations.

1.03 SUBMITTALS

A. Furnish submittals as specified in Section 01330 - Submittal Procedures.

- ✓ B. Product data:
- ✓ 1. Manufacturer of wire and cable.
 - ✓ 2. Insulation:
 - ✓ a. Type.
 - ✓ b. Voltage class.
 - ✓ 3. AWG size.
 - ✓ 4. Conductor material.
 - ✓ 5. Pulling compounds.

C. Shop drawings:

1. Show splice locations.
 - a. For each proposed splice location provide written justification describing why the splice is necessary.

D. Test reports:

1. Submit test reports for meg-ohm tests.

E. Calculations:

1. Submit cable pulling calculations to the Engineer for review and comment for all cables that will be installed using mechanical pulling equipment. Show that the maximum cable tension and sidewall pressure will not exceed manufacturer recommended values:
 - a. Provide a table showing the manufacturer's recommended maximum cable tension and sidewall pressure for each cable type and size included in the calculations.
 - b. Submit the calculations to the Engineer a minimum of 2 weeks before conduit installation.

F. Cable lengths:

1. Submit installed cable lengths using a conduit measuring tape for all 3-phase circuits.

2. Submit installed lengths of cable for the following single-phase circuits:
 - a. Circuits feeding single-phase transformers.
 - b. Circuits feeding single-phase panelboards.

1.04 QUALITY ASSURANCE

- A. All wires and cables shall be UL listed and labeled.

1.05 PROJECT OR SITE CONDITIONS

- A. General site and project conditions:
 1. As specified in Section 01850 - Design Criteria.

1.06 WARRANTY

- A. As specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 GENERAL

- A. Color-coding:
 1. Color-coding shall be consistent throughout the facility.
 2. The following color code shall be followed for all 240/120 volt and 208/120-volt systems:
 - a. Phase A: Black.
 - b. Phase B: Red.
 - c. Phase C: Blue.
 - d. Single phase system: Black for 1 hot leg, red for the other.
 - e. Neutral: White.
 - f. High phase or wild leg: Orange.
 - g. Equipment ground: Green.
 3. The following color code shall be followed for all 480/277-volt systems:
 - a. Phase A: Brown.
 - b. Phase B: Orange.
 - c. Phase C: Yellow.
 - d. Neutral: Gray.
 - e. Equipment ground: Green.
 4. The following color code shall be followed for all 120 VAC control wiring:
 - a. Power: Red.
 - b. Neutral: White.
 5. The following color code shall be followed for all general purpose DC control circuits:
 - a. Grounded conductors: White with blue stripe.
 - b. Ungrounded conductors: Blue.
 6. Switch legs shall be violet. 3-way switch runners shall be pink.
 7. Wires in intrinsically safe circuits shall be light blue.
 8. Wire colors shall be implemented in the following methods:
 - a. Wires manufactured of the desired color.

- b. Continuously spiral wrap the first 6 inches of the wire from the termination point with colored tape:
 - 1) Colored tape shall be wrapped to overlap 1/2 of the width of the tape.

2.02 DESIGN AND PERFORMANCE CRITERIA

- A. Furnish and install the complete wire and cable system.

2.03 MANUFACTURERS

- A. One of the following or equal:
 - 1. 600-volt class wire and cable:
 - a. Okonite Co.
 - b. Prysmian General Cable.
 - c. Southwire Co.
 - d. Service Wire.
 - 2. 600-volt VFD cable:
 - a. Prysmian General Cable.
 - b. Southwire Co.
 - c. Service Wire.
 - 3. Instrumentation class wire and cable:
 - a. Alpha Wire Co.
 - b. Belden CDT.
 - c. Okonite Co.
 - d. Prysmian General Cable.
 - e. Rockbestos Surprenant Cable Corp.
 - 4. Network cables:
 - a. Belden.
 - b. CommScope.
 - c. General Cable

2.04 MATERIALS

- A. Conductors:
 - 1. Copper in accordance with ASTM B3.

2.05 MANUFACTURED UNITS

- ✓ A. General:
 - ✓ 1. Provide new wires and cables manufactured within 1 year of the date of delivery to the Site.
 - ✓ 2. Permanently mark each wire and cable with the following at 24-inch intervals:
 - ✓ a. AWG size.
 - ✓ b. Voltage rating.
 - ✓ c. Insulation type.
 - ✓ d. UL symbol.
 - ✓ e. Month and year of manufacture.
 - ✓ f. Manufacturer's name.
 - ✓ 3. Identify and mark wire and cable as specified in Section 16075 - Identification for Electrical Systems:
 - ✓ a. Use integral color insulation for #2 AWG and smaller wire.

- ✓ b. Wrap colored tape around cable larger than #2 AWG.
- ✓ B. 600-volt class wire and cable:
 - ✓ 1. Provide AWG or kcmil sizes as indicated on the Drawings or in the Conduit Schedules:
 - ✓ a. When not indicated on the Drawings, size wire as follows:
 - ✓ 1) In accordance with the NEC:
 - ✓ a) Use 75-degree Celsius ampacity ratings.
 - ✓ b) Ampacity rating after all derating factors, equal to or greater than rating of the overcurrent device.
 - ✓ 2) Provide #12 AWG minimum for power conductors.
 - ✓ 3) Provide #14 AWG minimum for control conductors.
 - ✓ 2. Provide Class B stranding in accordance with ASTM B8:
 - ✓ a. Provide Class C stranding where extra flexibility is required.
 - ✓ 3. Insulation:
 - ✓ a. XHHW-2.
 - ✓ b. 90-degree Celsius rating.
- ✓ C. 600-volt VFD cables:
 - ✓ 1. Conductor:
 - ✓ a. Provide Class B stranding in accordance with ASTM B3 and B8.
 - ✓ b. Copper conductor.
 - ✓ 2. Insulation:
 - ✓ a. Cross-linked Polyethylene, XLPE.
 - ✓ b. 90-degree Celsius rating.
 - ✓ c. UL 44 Type RHH/RHW-2 or XHHW-2.
 - ✓ d. 2,000 volts.
 - ✓ 3. Ground:
 - ✓ a. Provide Class B compressed stranded bare copper in accordance with ASTM B3 and B8.
 - ✓ b. 3 symmetrically placed bare copper conductors in direct contact with shield.
 - ✓ c. Equivalent ground size 50 percent of the phase conductor size.
 - ✓ 4. Filler:
 - ✓ a. Cable sizes 8 and 6 use polypropylene filler.
 - ✓ b. Other cable sizes use paper filler.
 - ✓ 5. Metallic shield:
 - ✓ a. Overall 5 mil bare copper tape shield with 50 percent overlap.
 - ✓ 6. Outer jacket:
 - ✓ a. UL 1277 Type PVC.
 - ✓ 7. Cable tray rated, UL Type TC-ER.
- ✓ D. Instrumentation class cable:
 - ✓ 1. Type TC.
 - ✓ 2. Suitable for use in wet locations.
 - ✓ 3. Voltage rating: 600 volts.
 - ✓ 4. Temperature rating:
 - ✓ a. 90-degree Celsius rating in dry locations.
 - ✓ b. 75-degree Celsius rating in wet locations.

- ✓ 5. Conductors:
 - ✓ a. Insulation:
 - ✓ 1) Flame-retardant PVC, 15 mils nominal thickness, with nylon jacket 4 mils nominal thickness.
 - ✓ b. #16 AWG stranded and tinned.
 - ✓ c. Color code: ICEA Method 1:
 - ✓ 1) Pair: Black and white.
 - ✓ 2) Triad: Black, white and red.
 - ✓ 3) Multiple pairs or triads:
 - ✓ a) Color-coded and numbered.
- ✓ 6. Drain wire:
 - ✓ a. #18 AWG.
 - ✓ b. Stranded, tinned.
- ✓ 7. Jacket:
 - ✓ a. Flame retardant, moisture and sunlight resistant PVC.
 - ✓ b. Ripcord laid longitudinally under jacket to facilitate removal.
- ✓ 8. Shielding:
 - ✓ a. Individual pair/triad:
 - ✓ 1) Minimum 1.35-mil double-faced aluminum foil-polyester tape overlapped to provide 100 percent coverage.
 - ✓ b. Multiple pair or triad shielding:
 - ✓ 1) Group shield: Minimum 1.35-mil double-faced aluminum foil-polyester tape overlapped to provide 100 percent coverage.
 - ✓ 2) Completely isolate group shields from each other.
 - ✓ 3) Cable shield: 2.35 mils double-faced aluminum and synthetic polymer backed tape overlapped to provide 100 percent coverage.
 - ✓ c. All shielding to be in contact with the drain wire.
- ✓ E. Network cables:
 - ✓ 1. Copper Ethernet cable:
 - ✓ a. Provide copper Ethernet cable types as indicated in the Drawings and Specifications.
 - ✓ 2. The following color code shall be followed for all Ethernet cables:
 - ✓ a. IT/Business Network: Blue.
 - ✓ b. HMI Network: Yellow.
 - ✓ c. PLC Network: Green.
 - ✓ d. PLC Field Network: Black.
 - ✓ e. Security/Fire: Red.
 - ✓ f. General requirements:
 - ✓ 1) All cables shall meet the standards set by TIA-568.2-D and verified by third-party testing laboratory.
 - ✓ 2) Conductors:
 - ✓ a) Four balanced twisted pairs.
 - ✓ (1) #22 to #24 AWG thermoplastic insulated solid copper conducts enclosed by a thermoplastic jacket. Copper clad aluminum is not allowed.
 - ✓ 3) Insulation:
 - ✓ a) Non-Plenum: Polyolefin.
 - ✓ b) Plenum: Fluoropolymer.
 - ✓ 4) Strand color coded per T568B.
 - ✓ 5) Outer jacket with ripcord.

- ✓ 6) Shielding:
 - ✓ a) Provide F/UTP cables with drain wire for cables inside any equipment or enclosure and as indicated on the Drawings.
- ✓ 7) Voltage rating:
 - ✓ a) At a minimum provide 300 VAC rated jacket.
 - b) 600 VAC rated jacket when cables are inside equipment or enclosures that contain 480 VAC power.
- ✓ 8) Approvals and listings:
 - ✓ a) Meets any necessary NEC and NFPA requirements for each application.
 - ✓ b) Riser applications: CMR in accordance with UL 1666.
 - ✓ c) Plenum applications: CMP in accordance with NFPA 262.
- ✓ 9) Certification:
 - ✓ a) Provide Category 6 cables with NEMA WC 66/ICEA S-116-732 certification.
- 3. RS-485 cable:
 - a. 4-wire:
 - ✓ 1) Shielded twisted pair.
 - ✓ 2) Tinned copper conductors minimum with 7 by 30 stranding.
 - ✓ 3) #22 AWG.
 - ✓ 4) Insulation:
 - ✓ a) FHDPE.
 - ✓ b) 300-volt insulation level.
 - ✓ 5) Outer shield:
 - ✓ a) 100 percent coverage.
 - ✓ b) Tape/braid.
 - ✓ c) Aluminum foil-polyester tape.
 - ✓ d) Tinned copper braid.
 - ✓ 6) Outer shield drain wire:
 - ✓ a) Tinned copper conductor minimum with 7 by 30 stranding.
 - ✓ b) #22 AWG.
 - ✓ 7) Outer jacket PVC:
 - ✓ a) Sunlight resistant.
 - ✓ 8) UL/CSA flame tested.
 - ✓ 9) Minimum bending radius 2.5 inches.
 - 10) Nominal OD 0.284 inch.
 - ✓ 11) Outer shield drain wire:
 - ✓ a) Tinned copper conductor minimum with 7 by 30 stranding.
 - ✓ b) #22 AWG.
 - ✓ 12) Outer jacket PVC.
 - ✓ a) Sunlight resistant.

2.06 ACCESSORIES

- ✓ A. Wire ties:
 - ✓ 1. One of the following or equal:
 - ✓ a. T&B, "Ty-Rap" cable ties.
 - ✓ b. Panduit, cable ties.
- B. Wire markers:
 - 1. As specified in Section 16075 - Identification for Electrical Systems.

2.07 SOURCE QUALITY CONTROL

- A. Assembly and testing of cable shall comply with the applicable requirements of NEMA WC 70/ICEA S-95-658-1999.
- B. Test Type XHHW-2 in accordance with the requirements of UL 44.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install conductors only after the conduit installation is complete, and all enclosures have been vacuumed clean, and the affected conduits have been swabbed clean and dry:
 - 1. Install wires only in approved raceways.
 - 2. Do not install wire:
 - a. In incomplete conduit runs.
 - b. Until after the concrete work and plastering is completed.
- B. Properly coat wires and cables with pulling compound before pulling into conduits:
 - 1. For all #4 AWG and larger, use an approved wire-pulling lubricant while cable is being installed in conduit:
 - a. Ideal Products.
 - b. Polywater Products.
 - c. 3M Products.
 - d. Greenlee Products.
 - e. Or equal as recommended by cable manufacturer.
 - f. Do not use oil, grease, or similar substances.
- C. Cable pulling:
 - 1. Prevent mechanical damage to conductors during installation.
 - 2. For cables #1 AWG and smaller, install cables by hand.
 - 3. For cables larger than #1 AWG, power pulling winches may be used if they have cable tension monitoring equipment.
 - 4. Provide documentation that maximum cable pulling tension was no more than 75 percent of the maximum recommended level as published by the cable manufacturer. If exceeded, the Engineer may, at his discretion, require replacement of the cable.
 - 5. Ensure cable pulling crews have all calculations and cable pulling limitations while pulling cable.
 - 6. Make splices or add a junction box or pullbox where required to prevent cable pulling tension or sidewall pressure from exceeding 75 percent of manufacturer's recommendation for the specified cable size:
 - a. Make splices in manholes or pull boxes only.
 - b. Leave sufficient slack to make proper connections.
- D. Install and terminate all wire in accordance with manufacturer's recommendations.

- E. Neatly arrange and lace conductors in all switchboards, panelboards, pull boxes, and terminal cabinets by means of wire ties:
 - 1. Do not lace wires in gutter or panel channel.
 - 2. Install all wire ties with a flush cutting wire tie installation tool:
 - a. Use a tool with an adjustable tension setting.
 - 3. Do not leave sharp edges on wire ties.
- F. Terminate stranded conductors on equipment box lugs such that all conductor strands are confined within the lug:
 - 1. Use ring type lugs if box lugs are not available on the equipment.
- G. Lighting circuits:
 - 1. Each circuit shall have a dedicated neutral.
- H. Splices:
 - 1. Provide continuous circuits from origin to termination whenever possible:
 - a. Obtain Engineer's approval prior to making any splices.
 - 2. Lighting and receptacle circuit conductors may be spliced without prior approval from the Engineer.
 - 3. Where splices are necessary because of extremely long wire or cable lengths that exceed standard manufactured lengths:
 - a. Splice box NEMA rating requirements as specified in Section 16050 - Common Work Results for Electrical.
 - b. Make splices in labeled junction boxes for power conductors.
 - c. Make splices for control and instrument conductors in terminal boxes:
 - 1) Provide terminal boards with setscrew pressure connectors, with spade or ring lug connectors.
 - 4. Power and control conductors routed in common raceways may be spliced in common junction boxes.
 - 5. Clearly label junction and terminal boxes containing splices with the word "SPICE LOCATED WITHIN".
 - 6. Leave sufficient slack at junction boxes and termination boxes to make proper splices and connections. Do not pull splices into conduits.
 - 7. Install splices with compression type butt splices and insulate using a heat-shrink sleeve:
 - a. In NEMA Type 4 or NEMA Type 4X areas, provide heat-shrink sleeves that are listed for submersible applications.
 - 8. Splices in below grade pull boxes, in any box subject to flooding, and in wet areas shall be made waterproof using:
 - a. A heat shrink insulating system listed for submersible applications.
 - b. Or an epoxy resin splicing kit.
- I. Apply wire markers to all wires at each end after being installed in the conduit and before meg-ohm testing and termination.
- J. Instrumentation class cable:
 - 1. Install instrumentation class cables in separate raceway systems from power cables:
 - a. Install instrument cable in metallic conduit within non-dedicated manholes or pull boxes.

- b. Install cable without splices between instruments or between field devices and instrument enclosures or panels.
- 2. Do not make intermediate terminations, except in designated terminal boxes as indicated on the Drawings.
- 3. Shield grounding requirements as specified in Section 16060 - Grounding and Bonding.
- K. Copper Ethernet cables:
 - 1. In accordance with TIA-568.2-D.
 - 2. Pathways:
 - a. For initial installation, the maximum fill capacity for pathways (i.e., conduit, raceways, trays, baskets) is 40 percent. The maximum fill capacity of 60 percent is allowed to accommodate future additions after initial installation.
 - b. Conduit should be run in the most direct route possible with no more than two 90-degree bends between pull boxes and serve no more than 3 outlet boxes.
 - 3. Cable bend radius:
 - a. Proper cable bend radius control must be maintained throughout the pathways. The bend radius needs to be at a minimum 10 times the cable diameter.
 - 4. Cable pulling:
 - a. Provide cable pulling swivel system to prevent winding and tangling of rope and cables during pull.
 - b. The maximum pulling tension is not to exceed manufacturer recommendations. Cable installation should not in any way deform the cable jacket.
 - c. Do not install bruised, kinked, scored, deformed, or abraded cable. Do not splice cable between termination, tap, or junction points. Remove and discard cable if damaged during installation and replace it with new cable.
 - 5. Cable management:
 - a. Organize and manage cables for quick and easy moves, adds and changes.
 - 6. Cable termination:
 - a. Install equipment outlet connector hardware (e.g., RJ45, M12, etc.), and connect to field equipment outlet (e.g., instrument, VFD, actuator, etc.).
 - b. Use shielded connectors as required by the installation.
 - c. Coordinate cable termination at copper patch panels with ICSC and General Contractor.
 - 7. Testing:
 - a. Copper Ethernet cable testing requirements as specified in Section 16950 - Field Electrical Acceptance Tests.
 - 8. Separation from EMI sources:
 - a. Comply with TIA-569-B recommendations for separating unshielded copper voice and data communication cable from potential EMI sources, including electrical power lines and equipment.
 - b. Separation between open communications cables or cables in nonmetallic raceways and unshielded power conductors and electrical equipment shall be as follows:
 - 1) Electrical Equipment Rating Less Than 2 kVA: A minimum of 5 inches.

- 2) Electrical Equipment Rating between 2 and 5 kVA: A minimum of 12 inches.
 - 3) Electrical Equipment Rating More Than 5 kVA: A minimum of 24 inches
 - c. Separation between communications cables in grounded metallic raceways and unshielded power lines or electrical equipment shall be as follows:
 - 1) Electrical Equipment Rating Less Than 2 kVA: A minimum of 2-1/2 inches.
 - 2) Electrical Equipment Rating between 2 and 5 kVA: A minimum of 6 inches.
 - 3) Electrical Equipment Rating More Than 5 kVA: A minimum of 12 inches.
 - d. Separation between communications cables in grounded metallic raceways and power lines and electrical equipment located in grounded metallic conduits or enclosures shall be as follows:
 - 1) Electrical Equipment Rating Less Than 2 kVA: No requirement.
 - 2) Electrical Equipment Rating between 2 and 5 kVA: A minimum of 3 inches.
 - 3) Electrical Equipment Rating More Than 5 kVA: A minimum of 6 inches.
 - e. Separation between Communications Cables and Electrical Motors and Transformers, 5 kVA or HP and Larger: A minimum of 48 inches.
- L. Signal cable:
- 1. Separate and isolate electrical signal cables from sources of electrical noise and power cables by minimum 12 inches.
- M. Submersible cable in wet wells:
- 1. Provide Kellems grip or stainless steel wire mesh to support cable weight and avoid stress on insulation.
- N. Wiring allowances:
- 1. Equipment locations may vary slightly from the drawings. Include an allowance for necessary conductors and terminations for motorized equipment, electrical outlets, fixtures, communication outlets, instruments, and devices within 10 linear feet of locations indicated on the Drawings.
 - 2. Locations for pull boxes, manholes, and duct banks may vary slightly from the drawings. Include an allowance for necessary conductors and related materials to provide conductors to all pull boxes, manholes and duct banks within 20 linear feet of locations indicated on the Drawings.

3.02 COMMISSIONING

- A. As specified in Section 01756 - Commissioning.
- B. Functional Testing:
 - 1. As specified in Section 16950 - Field Electrical Acceptance Tests.

END OF SECTION

1.0 600-VOLT CLASS WIRE AND CABLE

SPECIFICATION – 16123-2.03.A.1 600-VOLT OR LESS WIRES AND CABLES





16123-2.05.B.3.a

SIMPull XHHW-2[®]/RW90 Copper

Single Conductor Copper, Cross Linked Polyethylene (XLPE) with SIMpull technology for easier pulling. Type XHHW-2 Conductor sizes 8 AWG and larger are rated 600 or 1000 Volt. Type RW90 Conductors are rated for 600 Volt.



See Table 3 For Other Color Options



Image not to scale. See Table 1 for dimensions.

CONSTRUCTION:

16123-2.05.B.2

1. **Conductor:** Class B compressed stranded bare copper per ASTM B3 and ASTM B8.
2. **Insulation:** Cross Linked Polyethylene (XLPE) with SIMpull[®] Technology. Silicone-Free, Abrasion, High-Heat, Moisture Resistant

16123-2.05.B.2

APPLICATIONS AND FEATURES:

APPLICATION

Southwire Copper SIMpull XHHW-2[®]/RW90 conductors are primarily used in conduit, cable tray or other recognized raceways for services, feeders, and branch circuit wiring. SIMpull XHHW-2[®]/RW90 copper conductors may be used in wet or dry locations at temperatures not to exceed 90°C. Southwire Copper SIMpull XHHW-2[®]/RW90 conductors are designed to be installed without the application of pulling lubricant.

Voltage rating for XHHW-2 is 600 volts or 1000 volts for sizes 8 and larger. Type XHHW-2 is suitable for use in Health Care Facilities per Section 517.160 of the National Electrical Code where a dielectric constant of less than 3.5 maybe specified.

Voltage rating for RW90 is 600 volts. RW90 is suitable for use in open wiring and use in raceways (except cable troughs and ventilated flexible cableways) in dry or wet locations as per Canadian Electrical Code. For open wiring exposed to the weather.

FEATURES

- SIS- 8 AWG
- Sunlight resistant
- -40°C Cold bend
- FT1
- Gasoline and Oil Resistant II
- CT Rated- 1/0 AWG and larger
- FT4- 350 kcmil and larger
- RoHS/REACH Compliant
- SPEC 10005 for circuit sizes 14 AWG, 12 AWG, and 10 AWG

SPECIFICATIONS:

- ASTM B3 Soft or Annealed Copper Wire

16123-2.04.A.1





- ASTM B8 Concentric-Lay-Stranded Copper Conductors
- ASTM B787 19 Wire Combination Unilay-Stranded Copper Conductors
- UL 44 Thermoset-Insulated Wires and Cables
- UL 1685 Vertical-Tray Fire Propagation and Smoke Release Test (1/0 and Larger)
- CSA C22.2 No. 38 Thermoset-insulated wires and cables
- ICEA S-95-658 (NEMA WC70) Power Cables Rated 2000 Volts or Less for the Distribution of Electrical Energy
- IEEE 1202/FT4 Flame Test (70,000 BTU/hr) 350kcmil and Larger
- Made in America: Compliant with both Buy American and Buy America Act (BAA) requirements per 49 U.S.C. § 5323(j) and the Federal Transit Administration Buy America requirements per 49 C.F.R. part 661
- Federal Specification A-A-59544
- NMX-J-451-ANCE Thermoset insulated wires and cables
- NOM-063-SCFI Electrical Products – Conductors – Safety Requirements
- NEMA 70901-2-2024 Make It American Compliance with Domestic Preference Requirements Pt. 2 Wire & Cable

SAMPLE PRINT LEGEND:

8 AWG thru 1 AWG

{SQFTG} SOUTHWIRE{R} {NOLUBE}{R} {SIMPULL}{R} E30117 {UL} TYPE XHHW-2 8 AWG (8.37{MM2}) CU 600V/1000V SR GRII PRII OR SIS 600V - LL90458 {CSA} RW90 XLPE 8 AWG (8.37{mm2}) CU 600V GRI PRI -40{D}C SR FT1 - {NOM}-ANCE LS - PAT WWW.PATENTSW.COM

1/0 AWG thru 300 kcmil

{SQFTG} SOUTHWIRE{R} {NOLUBE}{R} {SIMPULL}{R} E30117 {UL} TYPE XHHW-2 1/0 AWG (53.5{MM2}) CU 600V/1000V SR FOR CT USE GRII PRII - LL90458 {CSA} RW90 XLPE 1/0 AWG (53.5{mm2}) CU 600V GRI PRI -40{D}C SR FT1 - {NOM}-ANCE LS - PAT WWW.PATENTSW.COM

350 kcmil and Larger

{SQFTG} SOUTHWIRE{R} {NOLUBE}{R} {SIMPULL}{R} E30117 {UL} TYPE XHHW-2 350 KCMIL (177{MM2}) CU 600V/1000V SR FOR CT USE GRII PRII FT4 - LL90458 {CSA} RW90 XLPE 350 KCMIL (177{mm2}) CU 600V GRI PRI -40{D}C SR FT4 - {NOM}-ANCE LS - PAT WWW.PATENTSW.COM

16123-2.0.5.A.2





Table 1 – Weights and Measurements

Cond. Size AWG/Kcmil	Strand Count No. of Strands	Diameter Over Conductor inch	Insul. Thickness mil	Insulation Color	Approx. OD inch	Copper Weight lb/1000ft	Approx. Weight lb/1000ft
8	7	0.141	45	RD	0.238	50	64
6	7	0.177	45	BK	0.274	81	97
4	7	0.225	45	BK	0.321	128	148
3	7	0.252	45	BK	0.348	162	184
2	7	0.282	45	BK	0.379	204	228
1	19	0.322	55	OE	0.435	258	291
1/0	19	0.361	55	PE	0.476	326	361
2/0	19	0.405	55	PE	0.520	410	450
3/0	19	0.456	55	PE	0.570	518	561
4/0	19	0.512	55	PE	0.626	655	704
250	37	0.558	65	PE	0.696	771	834
300	37	0.610	65	GN	0.732	926	992
350	37	0.661	65	PE	0.799	1080	1154
400	37	0.705	65	BE	0.848	1235	1317
500	37	0.789	65	BE	0.908	1543	1632
600	61	0.865	80	PE	1.008	1852	1964
700	61	0.935	80	BN	1.103	2161	2284
750	61	0.960	80	BE	1.107	2315	2433
1000	61	1.117	80	OE	1.205	3087	3230

All dimensions are nominal and subject to normal manufacturing tolerances

◊ Cable marked with this symbol is a standard stock item





Table 2 – Electrical and Engineering Data

Cond. Size	Min Bending Radius	Max Pull Tension	DC Resistance @ 25°C	AC Resistance @ 75°C	Inductive Reactance @ 60Hz	Allowable Ampacity At 75°C	Allowable Ampacity At 90°C
AWG/Kcmil	inch	lb	Ω/1000ft	Ω/1000ft	Ω/1000ft	Amp	Amp
8	1.0	132	0.653	0.786	0.052	50	55
6	1.1	209	0.411	0.495	0.051	65	75
4	1.3	333	0.258	0.310	0.048	85	95
3	1.4	420	0.205	0.246	0.047	100	115
2	1.5	530	0.162	0.195	0.045	115	130
1	1.7	669	0.128	0.154	0.046	130	145
1/0	1.9	844	0.102	0.122	0.044	150	170
2/0	2.1	1064	0.081	0.097	0.043	175	195
3/0	2.3	1342	0.064	0.078	0.042	200	225
4/0	2.5	1692	0.051	0.062	0.041	230	260
250	2.8	2000	0.043	0.053	0.041	255	290
300	2.9	2400	0.036	0.045	0.041	285	320
350	3.2	2800	0.031	0.039	0.040	310	350
400	3.4	3200	0.027	0.035	0.040	335	380
500	3.6	4000	0.022	0.029	0.039	380	430
600	5.0	4800	0.018	0.025	0.039	420	475
700	5.5	5600	0.015	0.022	0.038	460	520
750	5.5	6000	0.014	0.022	0.038	475	535
1000	6.4	8000	0.011	0.018	0.037	545	615

* Ampacities based upon 2023 NEC Table 310.16 and do not consider the overcurrent protection limitations in NEC 240.4(D) of 15 Amps for 14 AWG CU, 20 Amps for 12 AWG CU, and 30 Amps for 10 AWG CU (independent of the conductor temperature rating and stranding if size is present in table). Also, see NEC sections 310.15 and 110.14(C) for additional requirements.

* Inductive impedance is based on non-ferrous conduit with one diameter spacing center-to-center.

* Ampacities derived from the 2021 Canadian Electrical Code. - Table 2 - for Raceway or Cable. Not more than 3 copper conductors on an ambient temperature of 30°C.





Table 3 - Stock Code Colors

Size (Strand)	Black	Red	Blue	White	Brown	Orange	Yellow	Gray	Purple	Green
8 (7)	112953	952713	553059	952721	553060	553061	553062	553063		952739
6 (7)	112961	952705	959916	678607	683383	683391	553067	553068		553230
4 (7)	112979	952697	553846	678599	553847	553848	553849	553850		558627
3 (7)	267278	652971	652972	652973	677646	677647	652975	677648		890469
2 (7)	112987	218115	553087	218107	553088	553089	553090		674066	474122
1 (19)	112995	550761	550762	550808	553854	553855	553856	553857	674065	550766
1/0 (19)	113001	553860	553861	553858	553863	553864	553865	553866	674064	553862
2/0 (19)	113019	553871	553872	553870	553873	553874	553876	553877	553878	552070
3/0 (19)	113027	553881	553882	553880	553885	553886	553887	553888	674063	553884
4/0 (19)	113035	553078	553079	553077	553080	553082	553083	553084	674062	552071
250 (37)	113043	553893	553894	553892	553896	553897	553898	553899	674061	553895
300 (37)	113050	643848	643849	643850	561129	561130	561131	584039	139410	580121
350 (37)	113068	553903	553904	553902	553906	553907	553908	553910	674060	553905
400 (37)	113076	561701	561702	561132	561111	561112	561113	561703		558666
500 (37)	113084	550369	550261	553071	550260	550262	550259	553074	553075	553072
600 (61)	113092	553914	553915	553913	553918	553919	553920	553921	589214	553916
700 (61)	586272				662040	662042	662043	662044		
750 (61)	113100	553927	553928	553920	553930	553931	553932	553934		553929
1000 (61)	113134	138623	138622	138624	670463	670465	670464			660449

Award Winning Patent
Pending Building Wire
Selector



2.0 600-VOLT VFD CABLE

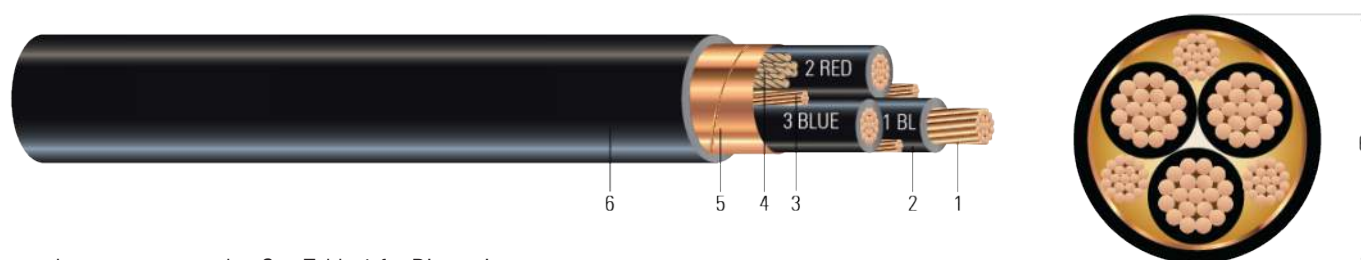
SPECIFICATION – 16123–2.03.A.2 600-VOLT OR LESS WIRES AND CABLES



3/C CU 2000V XLPE RHH/RHW-2 PVC VFD Power Cable Type TC-ER

Type TC-ER VFD Power Cable 2000 Volt Three Conductor Copper, Cross Linked Polyethylene (XLPE) insulation RHH/RHW-2 Polyvinyl Chloride (PVC) Jacket with 3 Symmetrical Bare CU Ground 50% Minimum Tape Shield Overlap

16123-2.05.C.3.b



Images not to scale. See Table 1 for Dimensions

CONSTRUCTION:

1. **Conductor:** Class B compressed stranded bare copper per ASTM B3 and ASTM B8 16123-2.05.C.1
2. **Insulation:** Cross Linked Polyethylene (XLPE) Type RHH/RHW-2 16123-2.05.C.2
3. **Grounding Conductor:** 3 Class B compressed stranded bare copper ground per ASTM B3 and ASTM B8 (ground size is 50% of the phase conductor) 16123-2.05.C.3
4. **Filler:** Paper filler (cable size 8 & 6 uses Polypropylene filler) 16123-2.05.C.4
5. **Tape Shield:** 5 mil copper tape shield with a minimum of 50% overlap 16123-2.05.C.5
6. **Overall Jacket:** Polyvinyl Chloride (PVC) Jacket 16123-2.05.C.2.d

16123-2.05.C.7

APPLICATIONS AND FEATURES:

Southwire's 2000 Volt Type TC-ER VFD power cables are suited for use in wet and dry areas, conduits, ducts, troughs, trays, direct burial, aerial supported by a messenger, and where superior electrical properties are desired. These cables are capable of operating continuously at the conductor temperature not in excess of 90°C for normal operation in wet and dry locations, 130°C for emergency overload, and 250°C for short circuit conditions. For uses in Class I, II, and III, Division 2 hazardous locations per NEC Article 501 and 502. Constructions with 3 or more conductors are listed for exposed runs (TC-ER) per NEC 336.10

16123-2.05.C.2.b

SPECIFICATIONS:

- ASTM B3 Soft or Annealed Copper
- ASTM B8 Concentric-lay-standard copper
- UL 44 Thermoset Insulated wires And cables 16123-2.05.C.2.c
- UL 1277 Electrical Power And Control Cable 16123-2.05.C.6
- UL 1685 - Flame Test
- UL 1581 - Electrical Wires, Cables and Flexible Cords
- IEEE 1202/FT4 - Vertical Tray Flame Test (70,000 Btu/hr)
- ICEA S-58-679 - Control Cable Conductor Identification Method 3 (1-BLACK, 2-RED, 3-BLUE)
- ICEA S-95-658 NEMA WC70 - Power cables rated 2000 volts or less for the distribution of electrical energy

SAMPLE PRINT LEGEND:

SOUTHWIRE EXXXXX #P# (UL) [#AWG Or #kcmil] CU RHH/RHW-2 XLPE/PVC 2000V Type TC-ER For CT USE SUN. RES. For DIRECT BURIAL FT4 YEAR (NESC) [SEQUENTIAL FEET MARKS]

16123-2.05.A.2



Southwire Company, LLC | One Southwire Drive, Carrollton, GA 30119 | www.southwire.com

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Southwire®

SPEC 45451_PSS DIVISION DATE: 03/27/2019 Rev:2.1.18M

Table 1 – Weights & Measurements

Stock Code	Cond. Size AWG	Dia Over Cond. (1) inches	Insul. Thickness inches	Dia Over Insul. (2) inches	Ground No. x AWG	Jacket Thickness mills	Approx. OD (6) inches	Copper Weight lbs./MFT	Approx. Weight lbs./MFT
580672 [◇]	14	0.070	60	0.190	3 x 18	45	0.519	54	175
580685 [◇]	12	0.087	60	0.207	3 x 16	45	0.558	85	220
580693 [◇]	10	0.111	60	0.231	3 x 14	60	0.638	136	307
569388 [◇]	8	0.139	70	0.279	3 x 14	60	0.743	193	427
580701 [◇]	6	0.174	70	0.314	3 x 12	60	0.819	307	586
569389 [◇]	4	0.221	70	0.361	3 x 12	80	0.959	452	789
569387 [◇]	2	0.277	70	0.417	3 x 10	80	1.081	718	1120
TDA	1	0.321	90	0.501	3 x 8	80	1.262	937	1460
644333 [◇]	1/0	0.360	90	0.540	3 x 6	80	1.346	1233	1805
644334 [◇]	2/0	0.404	90	0.584	3 x 6	80	1.441	1491	2121
644337	3/0	0.454	90	0.634	3 x 5	80	1.549	1880	2577
644338 [◇]	4/0	0.510	90	0.690	3 x 4	80	1.67	2370	3146
644339 [◇]	250	0.558	105	0.768	3 x 2	110	1.899	2960	3978
644340 [◇]	350	0.661	105	0.871	3 x 2	110	2.121	3896	5084
644341 [◇]	500	0.789	105	0.999	3 x 1	110	2.398	5461	6867

All dimensions are nominal and subject to normal manufacturing tolerances

◇ Standard stock item

Table 2 – Electrical and Engineering Data

Stock Code	Cond. Size AWG	Min. Bending Radius Inches	Max. Pull Tension lbs.	Resistance		Reactance X _L @ 60Hz Ω/MFT	Ø Short Circuit Current 6 Cycles Amps	Allowable Ampacities [†]		
				DC @ 25°C Ω/MFT	AC @ 90°C Ω/MFT			60 °C Amps	75 °C Amps	90 °C Amps
580672	14	6.2	99	2.630	3.288	0.045	935	15	15	15
580685	12	6.7	157	1.660	2.075	0.042	1485	20	20	20
580693	10	7.7	249	1.040	1.300	0.039	2360	30	30	30
569388	8	8.9	396	0.652	0.815	0.038	3754	40	50	55
580701	6	9.8	630	0.411	0.514	0.035	5966	55	65	75
569389	4	11.5	1002	0.258	0.323	0.033	9491	70	85	95
569387	2	13.0	1593	0.162	0.203	0.031	15089	95	115	130
TDA	1	15.1	2009	0.129	0.161	0.032	19029	110	130	145
644333	1/0	16.2	2534	0.102	0.128	0.031	24011	125	150	170
644334	2/0	17.3	3194	0.081	0.102	0.030	30264	145	175	195
644337	3/0	18.6	4027	0.064	0.081	0.029	38154	185	200	225
644338	4/0	20.0	5078	0.051	0.064	0.029	48114	195	230	260
644339	250	22.8	6000	0.043	0.055	0.029	56845	215	255	290
644340	350	25.5	8400	0.031	0.040	0.028	79583	280	310	350
644341	500	28.8	12000	0.022	0.028	0.027	113690	320	380	430

[†] Ampacities are based on Table 310.15 (B)(16) of the NEC, 2014 Edition. Ampacities of insulated conductors rated up to and including 2000 Volts, based on ambient temperature of 30°C (86°F)



3.0 INSTRUMENTATION CLASS WIRE AND CABLE

SPECIFICATION – 16123-2.03.A.3 600-VOLT OR LESS WIRES AND CABLES



UL Instrumentation Cable

600V Tray Cables – Overview

16123-2.05.D.5.b

16123-2.05.D.1

16123-2.05.D.7.a

Tray Cable Construction Options

UL Listed for MC and TC				
Insulation/Jacket	Max. Temp Rating		Flame Tests	Ratings*
	Wet	Dry		
PVC-Nylon/PVC (THHN or THWN) 14 AWG & larger	75°C	90°C	UL 1685 FT4/IEEE 1202/383 ICEA T-29-520	ICEA S-73-532 ICEA S-61-402
PVC-Nylon/PVC (TFN or TFFN) 16 & 18 AWG	N/A	90°C	UL 1685 FT4/IEEE 1202/383 ICEA T-29-520	ICEA S-73-532 ICEA S-61-402
XLPE/PVC or CPE (XHHW-2) 14 AWG & larger	90°C	90°C	UL 1685 FT4/IEEE 1202/383 VW-1 rated singles ICEA T-29-520	ICEA S-73-532 ICEA S-66-524
XLPE/PVC or CPE (RFH-2) 16 & 18 AWG	75°C	75°C	UL 1685 FT4/IEEE 1202/383 VW-1 rated singles ICEA T-29-520	ICEA S-73-532 ICEA S-66-524 ICEA S-82-552
FRPO/PVC 18 AWG & larger	—	75°C	UL 1685	
TPE/TPE	75°C	90°C	UL 1685	
FRPO/PVC	75°C	90°C	UL 1685	
XLPE/Haloarrest® (XHHW-2) 14 AWG & larger	90°C	90°C	UL 1685 ICEA T-29-520 FT4/IEEE 1202/383	TC-LS
XLPE/Haloarrest (RFH-2) 16 & 18 AWG	75°C	75°C	UL 1685 ICEA T-29-520 FT4/IEEE 1202/383	TC-LS
FEP/PVC	90°C	90°C	UL 1685	

CPE = Chlorinated Polyethylene • FEP = Fluorinated Ethylene-propylene • FRPO = Flame-retardant Polyolefin • PVC = Polyvinyl Chloride • TPE = Thermoplastic Elastomer • XLPE = Cross-linked Polyethylene
*Applicable to TC-rated cables only.

Construction

Soft annealed bare or tinned copper conductors. PVC insulated with a nylon overcoat, 90°C PVC Jacket, TFN, TFFN or THHN style singles. Nylon rip cord included in all PVC-Nylon/PVC instrumentation cables.

Application

These cables are suitable for installation in wet or dry locations. Cable jackets are resistant to sunlight, moisture and vapor penetration. The cables can be used in raceways, and (supported by messenger wire), outdoor applications and direct burial applications.

Unshielded

Twisted non-shielded instrument pairs provide a minimal OD allowing greater tray and conduit fill. Non-shielded instrument pairs may be utilized when recommended by the instrument manufacturer and used in a metallic conduit.

Overall Shield

Recommended for use in instrumentation applications where signals are transmitted in excess of 100 millivolts except in areas where high voltage and current sources creates excessive noise interference.

The Beldfoil® shield with drain wire provides 100% coverage for maximum shield effectiveness. Copper tape shield available upon request.

Individually Shielded and Overall Shielded

Individually shielded pairs or triads with an overall shield are recommended for use in instrumentation applications where optimum noise rejection is required. Individual pair/triad shields are fully isolated from each other and contain a separate drain wire for grounding, to provide maximum protection from crosstalk and common mode interference. Cables with an overall shield provide additional electrostatic noise protection.

Conductor, Insulation and Jacket Options*

To Specify:		Bare Tinned		Insulation/Jacket
1234	A	A	B	PVC-Nylon/PVC
		C	D	XLPE/PVC
E	F	G	H	FRPO/PVC
		I	J	XLPE/TPE
K	L	M	N	PVC-Nylon/Oil Res II
		O	P	XLPE/CPE
Q	R	S	T	XLPE/Haloarrest

*For 1000 and 3000 Series cables only.

Specifications

- UL Subject 1277 TC
- UL 1685 (UL 1581) Vertical Tray Flame Test comparable to IEEE 383-1974 (70,000 BTU/hr.) Flame Test
- NEC Type TC Listed, which is approved for cable tray use in Class 1, Division 2 areas, per NEC Articles 340, 318 and 501 and for Class 1 circuits as permitted in Article 725
- PVC-nylon/PVC constructions are NEC Type NPLF Listed, which is approved for use in Non Power-Limited Fire Protective Signaling circuits, per NEC Article 760
- PVC-Nylon/PVC, XLPE/PVC and XLPE/CPE constructed cables meet IEEE 1202/IEEE 383-2003/FT4 (70,000 BTU) Flame Test
- XLPE/Haloarrest cables are UL 1277 TC-LS rated

TC-ER Rated Cables

As an option, Belden offers all PVC-nylon/PVC, XLPE/PVC and XLPE/CPE jacketed tray cables with a TC-ER (Exposed Run) rating, formerly referred to as Open Wiring.

Per NEC Article 336, a TC-ER rated cable may be installed in an industrial establishment between a cable tray and the utilization equipment or device. A TC-ER rated cable must meet the crush and impact requirements of UL Type MC cable. By eliminating the need for metal conduit and/or armor, using a TC-ER rated cable results in savings in both installation and maintenance.

MC Cable Ratings Optional

Customize any 600V TC instrumentation cable, with armor and a full-sized ground. See chart below to specify.

To Specify MC Rated Cable			
1	2	3456	A
Overall Jacket Type	Armor Type	Core 4-digit Part No. 600V TC Instrumentation	Conductor, Insulation, Inner Jacket Type

Overall Jacket

Code	Material
1	PVC
2	CPE
3	TPE
4	HORE
5	Oil Res II
6	Haloarrest

Armor

Code	Material
2	Aluminum Interlock
3	Steel Interlock

Standard lengths may be subject to tolerance. Custom lengths may be available upon request. Contact the Belden Electronics Division Customer Service Department for additional information. 1-800-BELDEN-1.



UL Instrumentation Cable**600V Tray Cables**

16123-2.05.D.1

Industrial Grade Sunlight- and Oil-Resistant Jackets

16123-2.05.D.5.c

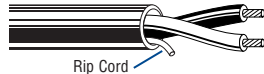
16123-2.05.D.7.a

Description	Part No.	No. of Pairs	Color Code	Standard Lengths		Standard Unit Weight		Jacket Thickness		Nominal OD		Maximum Pull Tension		Minimum Bend Radius	
				ft.	m	Lbs.	kg	Inch	mm	Inch	mm	Lbs.	N	Inch	mm

16 AWG Pairs Stranded (19x29) Tinned Copper Conductors • Twisted Pairs**Unshielded • PVC/Nylon Insulation • Black PVC Jacket**

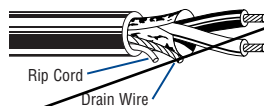
NEC: TC, NPLF
 ICEA S-73-532, ICEA S-61-402
 ICEA T-29-520
 FT4/IEEE 1202/383

9487	1	E2	500	152.4	25.5	11.6	.048	1.22	.295	7.49	70	311	3.00	76.20
			1000	304.8	54.0	24.5								

**Overall Beldfoil® Shield (100% Coverage) • PVC/Nylon Insulation • Black PVC Jacket**

NEC: TC, NPLF
 ICEA S-73-532, ICEA S-61-402
 ICEA T-29-520
 FT4/IEEE 1202/383

9342	1	E2	500	152.4	27.5	12.7	.048	1.22	.296	7.52	105	467	3.00	76.20
			1000	304.8	56.0	25.4								



16123-2.05.D.8.a.1

16 AWG Pairs Stranded (7x24) Bare Copper Conductors • Twisted Pairs**Overall Beldfoil Shield (100% Coverage) • PVC/Nylon Insulation • Black PVC Jacket** (See chart below for other options)

NEC: TC, NPLF
 ICEA S-73-532
 ICEA S-61-402
 ICEA T-29-520
 FT4/IEEE 1202/383

1118A	1	E2	10000 [†]	3048.0	550.0	249.5	.047	1.19	.294	7.47	105	467	3.00	76.20
3090A	1	E1	2500 [†]	762.0	150.0	68.0	.047	1.19	.295	7.49	105	467	3.00	76.20
1069A	2	E1	7500 [†]	2286.0	712.5	323.2	.047	1.19	.456	11.58	179	796	4.60	116.84
1527A	3	E1	7500 [†]	2286.0	1042.5	472.9	.047	1.19	.482	12.24	241	1072	4.80	121.92
1070A	4	E1	7500 [†]	2286.0	1357.5	615.8	.063	1.60	.560	14.22	321	1428	5.60	142.24
1871A	8	E1	7500 [†]	2286.0	2242.5	1017.2	.063	1.60	.676	17.17	607	2700	6.80	172.72
1072A	12	E1	5000 [†]	1524.0	1047.5	475.2	.063	1.60	.812	20.63	893	3972	8.10	205.74
1873A	16	E1	2500 [†]	762.0	1442.5	654.3	.085	2.16	.946	24.03	1178	5240	9.30	236.22
1074A	24	E1	1250 [†]	381.0	2115.0	959.4	.085	2.16	1.158	29.41	1749	7780	11.60	294.64
1089A	36	E1	1250 [†]	381.0	1388.7	629.9	.085	2.16	1.321	33.55	2606	11592	13.20	335.28
1890A	50	E1	Bulk ^{††}	Bulk			.085	2.16	1.551	39.40	3606	16040	15.50	393.70

Individually Shielded + Overall Beldfoil Shield (100% Coverage) • PVC/Nylon Insulation • Black PVC Jacket (Options below)

NEC: TC, NPLF
 ICEA S-73-532
 ICEA S-61-402
 ICEA T-29-520
 FT4/IEEE 1202/383

1055A	2	E1	7500 [†]	2286.0	885.0	401.8	.047	1.19	.476	12.09	223	992	4.16	105.66
1837A	3	E1	7500 [†]	2286.0	1072.5	486.5	.047	1.19	.504	12.80	290	1290	5.00	127.0
1039A	4	E1	7500 [†]	2286.0	1472.5	667.9	.063	1.60	.584	14.83	411	1828	5.80	147.32
1040A	6	E1	5000 [†]	1524.0	1435.0	650.9	.063	1.60	.682	17.32	428	1904	6.80	172.72
1841A	8	E1	5000 [†]	1524.0	1805.0	818.8	.063	1.60	.738	18.75	786	3496	7.40	187.96
1042A	12	E1	2500 [†]	762.0	1327.5	602.2	.085	2.16	.935	23.75	1161	5164	9.40	238.76
1843A	16	E1	2500 [†]	762.0	1765.0	800.6	.085	2.16	1.035	26.29	1537	6837	10.40	264.16
1844A	20	E1	2500 [†]	762.0	2062.5	935.6	.085	2.16	1.146	29.11	1912	8505	11.50	292.10
1045A	24	E1	1250 [†]	381.0	1241.3	563.1	.085	2.16	1.272	32.31	2287	10173	12.70	322.58
1046A	36	E1	Bulk ^{††}	Bulk			.085	2.16	1.454	36.93	3413	15182	14.50	368.30
1847A	50	E1	Bulk ^{††}	Bulk			.120	3.05	1.781	45.24	4726	21022	17.80	452.12

E1, E2 = Refer to Technical Information section for color codes.
 Alternate color coding available upon request.

Multiple pair or triad cables have each pair/triad numbered for ease of identification.

[†]Final put-up length may vary ±10% from length shown.
^{††}Bulk = Check length available for specific construction.

16123-2.05.D.5.c

16123-2.05.D.5.c

16123-2.05.D.8.c

To Specify:

1234	A	E
Start with Part No.	Add or replace letter code for desired conductor, insulation & jacket	Add for Exposed Run rating if desired

Conductor, Insulation and Jacket Options**

Bare	Tinned	Insulation/Jacket
A	B	PVC-Nylon/PVC
C	D	XLPE/PVC
E	F	FRPO/PVC
G	H	XLPE/TPE
K	L	TPE/TPE
M	N	PVC-Nylon/Oil Res II
Q	R	XLPE/CPE
S	T	XLPE/Haloarrest®

**For 1000 and 3000 Series cables only.

BELDEN

For more information, contact Belden Technical Support: 1-800-BELDEN-1 • www.belden.com

UL Instrumentation Cable**600V Tray Cables****Industrial Grade Sunlight- and Oil-Resistant Jackets**

16123-2.05.D.8.a.1

Description	Part No.	No. of Pairs/Triads	Color Code	Standard Lengths		Standard Unit Weight		Jacket Thickness		Nominal OD		Maximum Pull Tension		Minimum Bend Radius	
				Ft.	m	Lbs.	kg	Inch	mm	Inch	mm	Lbs.	N	Inch	mm

16 AWG Pairs Stranded (7x24) Bare Copper Conductors • Twisted Pairs**Overall Beldfoil Shield (100% Coverage) • PVC/Nylon Insulation • Black PVC Jacket (Green Ground)**

NEC: TC-ER, NPLF
ICEA S-73-532
ICEA S-61-402
ICEA T-29-520
FT4/IEEE 1202/383

3090AE

1

E1

1000

304.8

65.0

29.5

.048

1.22

.390

9.91

130

578

3.90

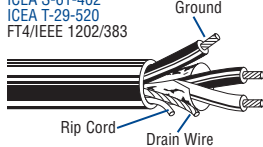
99.06

5000†

1524.0

340.0

154.4

**Overall Beldfoil Shield (100% Coverage) • XLPE Insulation • Black PVC Jacket (Green Ground)**

NEC: TC-ER, NPLF
ICEA S-73-532
ICEA S-66-524
ICEA T-29-520
FT4/IEEE 1202/383

3090CE

1

E1

1000

3048.0

81.0

39.8

.048

1.22

.390

9.91

130

578

3.90

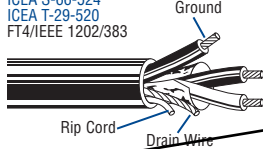
99.06

5000†

1524.0

450.0

204.3



16123-2.05.D.8.a.1

16 AWG Triads Stranded (7x24) Bare Copper Conductors • Twisted Triads**Overall Beldfoil® Shield (100% Coverage) • PVC/Nylon Insulation • Black PVC Jacket (See chart below for other options)**

NEC: TC, NPLF
ICEA S-73-532
ICEA S-61-402
ICEA T-29-520
FT4/IEEE 1202/383

1119A

1

E2

500†

152.4

35.0

15.9

.047

1.19

.310

7.87

129

574

3.10

78.74

1000†

304.8

68.0

30.9

10000†

3048.0

700.0

317.5

3091A

1

E1

10000†

3048.0

750.0

340.2

.047

1.19

.310

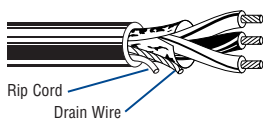
7.87

129

574

3.10

78.74



16123-2.05.D.8.b

Individually Shielded + Overall Beldfoil Shield (100% Coverage) • PVC/Nylon Insulation • Black PVC Jacket (Options below)

NEC: TC, NPLF
ICEA S-73-532
ICEA S-61-402
ICEA T-29-520
FT4/IEEE 1202/383

1097A

4

E1

5000†

1524.0

1415.0

641.8

.063

1.60

.640

16.26

554

2464

6.40

162.50

1098A

8

E1

2500†

762.0

1350.0

612.4

.085

2.16

.872

22.15

1072

4768

8.70

220.98

1099A

12

E1

2500†

762.0

1875.0

850.5

.085

2.16

1.047

26.59

1590

7073

10.50

266.70

3118A

16

E1

Bulk††

Bulk

Bulk

.084

2.13

1.234

31.34

1771

7878

12.25

311.15

1100A

24

E1

Bulk††

Bulk

Bulk

.085

2.16

1.434

36.42

3144

13985

14.30

363.22

3130A

36

E1

Bulk††

Bulk

Bulk

.110

2.79

1.773

45.03

3600

16013

18.00

457.20



E1, E2 = Refer to Technical Information section for color code.

Alternate color coding available upon request.

Multiple pair or triad cables have each pair/triad numbered for ease of identification.

†Final put-up length may vary ±10% from length shown.

††Bulk = Check length available for specific construction.

16123-2.05.D.5.c

16123-2.05.D.5.c

Conductor, Insulation and Jacket Options**To Specify:****1234**

Start with Part No.

A

Add or replace letter code for desired conductor, insulation & jacket

E

Add for Exposed Run rating if desired

Bare Tinned Insulation/Jacket**A B** **PVC-Nylon/PVC****C D** XLPE/PVC**E F** FRPO/PVC**G H** XLPE/TPE**K L** TPE/TPE**M N** PVC-Nylon/Oil Res II**Q R** XLPE/CPE**S T** XLPE/Haloarrest®



Product: [3090A](#) 

Inst, 1 Pr #16 Str BC, PVC-NYL Ins E1, OS, Blk PVC Jkt, 600V TC 150V NPLF 90C Dry/Wet

 [Request Sample](#)

16123-2.05.D.1

Product Description

UL Instrumentation, 1 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC 150V NPLF 90C Dry/Wet [SUN RES](#) DIR BUR

Technical Specifications

16123-2.05.D.7.a

Product Overview

Suitable Applications: Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners

Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	1	16 AWG	7x24	BC - Bare Copper

16123-2.05.D.5.b

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	Black/White

16123-2.05.D.5.c.1

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC

16123-2.05.D.5.a

16123-2.05.D.6.a

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.047 in (1.2 mm)	0.300 in (7.62 mm)	Yes

16123-2.05.D.8

16123-2.05.D.7.b

Overall Cable Diameter (Nominal): 0.300 in (7.62 mm)

16123-2.05.D.7.a

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Capacitance Cond-to-Cond	Nom. Capacitance Cond-to-Other (Conds + Shield)	Nom. Inductance	Max. Current
Pair(s)	4.45 Ohm/1000ft	50 pF/ft (160 pF/m)	87 pF/ft (290 pF/m)	0.091 µH/ft (0.30 µH/m)	6.2 Amps per conductor at 25°C

Nom Outer Shield DCR@Cond DCR1-cnvr: 5.47 Ohm/1000ft (17.9 Ohm/km)

Voltage

UL Voltage Rating
600 V (TC) 150 V (NPLF)

16123-2.05.D.3

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 90°C Wet	Wet: -30°C to +90°C, Dry: -25°C to +90°C

Bend Radius

16123-2.05.D.4

Stationary Min.	Installation Min.
3.0 in (76 mm)	3.0 in (76 mm)

Max. Pull Tension:	105 lbs (47.6 kg)
Bulk Cable Weight:	50 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 UL Loading, IEEE 1202, FT4
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC16123-2.05.D.5.a (7.a)
European Directive Compliance:	EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.610 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
3090A 0102500	Black	Reel	2,500 ft	612825141204	
3090A 0105000	Black	Reel	5,000 ft	612825141211	C Z
3090A 01010000	Black	Reel	10,000 ft	612825141198	C Z

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1069A](#) 

Inst, 2 Pr #16 Str BC, PVC-NYL Ins E1, OS, Blk PVC Jkt, 600V TC-ER 150V NPLF 90C Dry/Wet

 [Request Sample](#)

16123-2.05.D.1

Product Description

UL Instrumentation, 2 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V 150V NPLF 90C Dry/Wet

Technical Specifications

16123-2.05.D.7.a

Product Overview

Suitable Applications: Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners

Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	2	16 AWG	7x24	BC - Bare Copper

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	ICEA Chart E1

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.048 in (1.2 mm)	0.420 in (10.7 mm)	Yes

Overall Cable Diameter (Nominal): 0.420 in (10.7 mm)

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Capacitance Cond-to-Cond	Max. Current
Pair(s)	4.36 Ohm/1000ft	40 pF/ft (130 pF/m)	14 Amps per conductor @ 30°C

Nom Outer Shield DCR@Cond DCR1-cnvr: 3.24 Ohm/1000ft (10.6 Ohm/km)

Voltage

UL Voltage Rating
600 V (TC-ER), 150 V (NPLF)

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to 90°C Dry/Wet

Bend Radius

16123-2.05.D.4

Stationary Min.	Installation Min.
5.0 in (130 mm)	5.0 in (130 mm)

Max. Pull Tension:	179 lbs (81.2 kg)
Bulk Cable Weight:	94 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 FT4 Loading, FT4, IEEE 1202 FT4
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER16123-2.05.D.5.a (7.a)
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)
Other Standard Compliance(s):	07-KA080011-MSHA

History

Update and Revision:	Revision Number: 0.611 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1069A 0107500	Black	Reel	7,500 ft	612825105770	C Z

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Belden believes this product to be in compliance with all applicable environmental programs as listed in the data sheet. The information provided is correct to the best of Belden's knowledge, information and belief at the date of its publication. This information is designed only as a general guide for the safe handling, storage, and any other operation of the product itself or the one that it becomes a part of. The Product Disclosure is not to be considered a warranty or quality specification. Regulatory information is for guidance purposes only. Product users are responsible for determining the applicability of legislation and regulations based on their individual usage of the product.



Product: [1070A](#) 

Inst, 4 Pr #16 Str BC, PVC-NYL Ins E1, OS, Blk PVC Jkt, 600V TC-ER 150V NPLF 90C Dry/Wet

 [Request Sample](#)

16123-2.05.D.1

Product Description

UL Instrumentation, 4 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER 150V NPLF 90C Dry/Wet SUN RES DIR BUR

Technical Specifications

16123-2.05.D.7.a

Product Overview

Suitable Applications: Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners

Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	4	16 AWG	7x24	BC - Bare Copper

16123-2.05.D.5.b

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	ICEA Chart E1

16123-2.05.D.5.c.1

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

16123-2.05.D.5.a

16123-2.05.D.6.a

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.064 in (1.6 mm)	0.554 in (14.1 mm)	Yes

16123-2.05.D.8

16123-2.05.D.7.b

Overall Cable Diameter (Nominal): 0.554 in (14.1 mm)

16123-2.05.D.7.a

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Max. Current
Pair(s)	4.36 Ohm/1000ft	13 Amps per conductor @ 30°C

Nom Outer Shield DCR@Cond DCR1-cnvr: 3.14 Ohm/1000ft (10.3 Ohm/km)

Voltage

UL Voltage Rating
600 V (TC-ER) 150 V (NPLF)

16123-2.05.D.3

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to 90°C Dry/Wet

Bend Radius

16123-2.05.D.4

Stationary Min.	Installation Min.
6.6 in (170 mm)	6.6 in (170 mm)

Max. Pull Tension:	321 lbs (146 kg)
Bulk Cable Weight:	163 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 FT4 Loading, FT4, IEEE 1202 FT4
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER16123-2.05.D.5.a (7.a)
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.585 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1070A 0107500	Black	Reel	7,500 ft	612825105794	C Z

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1072A](#) 

Inst, 12 Pr #16 Str BC, PVC-NYL Ins E1, OS, Blk PVC Jkt, 600V TC-ER 150V NPLF 90C Dry 75C Wet

 [Request Sample](#)

16123-2.05.D.1

Product Description

UL Instrumentation, 12 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER 150V NPLF 90C Dry 75C Wet (SUN RES) DIR BUR

Technical Specifications

16123-2.05.D.7.a

Product Overview

Suitable Applications:	Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners
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Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	12	16 AWG	7x24	BC - Bare Copper

16123-2.05.D.5.b

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	Black, White and Numbered

16123-2.05.D.5.c.1

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

16123-2.05.D.5.a

16123-2.05.D.6.a

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.064 in (1.6 mm)	0.800 in (20.3 mm)	Yes

16123-2.05.D.8

16123-2.05.D.7.b

Overall Cable Diameter (Nominal):	0.800 in (20.3 mm)
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16123-2.05.D.7.a

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR
Pair(s)	3.67 Ohm/1000ft (12.0 Ohm/km)

Nom Outer Shield DCR@Cond DCR1-cnvr:	2.89 Ohm/1000ft (9.48 Ohm/km)
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Voltage

UL Voltage Rating
600 V (TC-ER) 150 V (NPLF)

16123-2.05.D.3

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 75°C Wet	-30C to 90C Dry, 75C wet

Bend Radius

16123-2.05.D.4

Stationary Min.	Installation Min.
9.6 in (240 mm)	9.6 in (240 mm)

Max. Pull Tension:	893 lbs (405 kg)
Bulk Cable Weight:	370.6 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL1685 (FT4 Loading), FT4, 1202
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER16123-2.05.D.5.a (7.a)
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.571 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1072A 0105000	Black	Reel	5,000 ft	612825105824	C Z

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1074A](#) 

Inst, 24 Pr #16 Str BC, PVC-NYL Ins E1, OS, Blk PVC Jkt, 600V TC-ER 150V NPLF 90C Dry/Wet

 [Request Sample](#)

16123-2.05.D.1

Product Description

UL Instrumentation, 24 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER 150V NPLF 90C Dry/Wet [SUN RES](#) DIR BUR

Technical Specifications

16123-2.05.D.7.a

Product Overview

Suitable Applications:	Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners
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Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	24	16 AWG	7x24	BC - Bare Copper

16123-2.05.D.5.b

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	ICEA Chart E1

16123-2.05.D.5.c.1

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

16123-2.05.D.5.a

16123-2.05.D.6.a

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.089 in (2.3 mm)	1.147 in (29.13 mm)	Yes

16123-2.05.D.7.b

Overall Cable Diameter (Nominal):	1.147 in (29.13 mm)
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16123-2.05.D.7.a

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Max. Current
Pair(s)	4.36 Ohm/1000ft	6 Amps per conductor @ 30°C

Nom Outer Shield DCR@Cond DCR1-cnvr:	2.69 Ohm/1000ft (8.83 Ohm/km)
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Voltage

UL Voltage Rating
600 V (TC-ER) 150 V (NPLF)

16123-2.05.D.3

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to 90°C Dry/Wet

Bend Radius

16123-2.05.D.4

Stationary Min.	Installation Min.
13.8 in (351 mm)	13.8 in (351 mm)

Max. Pull Tension:	1749 lbs (793.3 kg)
Bulk Cable Weight:	743 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 FT4 Loading, FT4, IEEE 1202 FT4
NEC / UL Compliance:	Article 336, Article 760, TC-ER, NPLF, TFFN 16123-2.05.D.5.a (7.a)
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.603 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1074A 0101250	Black	Reel	1,250 ft	612825105848	C Z
1074A 0105000	Black	Reel	5,000 ft	612825105855	
1074A 0107500	Black	Reel	7,500 ft	612825105862	

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1089A](#) 

Inst, 36 Pr #16 Str BC, PVC-NYL Ins E1, OS, Blk PVC Jkt, 600V TC-ER 150V NPLF 90C Dry 75C Wet

 [Request Sample](#)

16123-2.05.D.1

Product Description

UL Instrumentation, 36 Pair 16AWG (7x26) Bare Copper, PVC-NYL Insulation E1 Color Code, Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER 150V NPLF 90C Dry 75C Wet (SUN RES) DIR BUR

Technical Specifications

16123-2.05.D.7.a

Product Overview

Suitable Applications:	Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners
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Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	36	16 AWG	7x24	BC - Bare Copper

16123-2.05.D.5.b

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	Black, White and Numbered

16123-2.05.D.5.c.1

Outer Shield

16123-2.05.D.5.a

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

16123-2.05.D.6.a

Outer Jacket

16123-2.05.D.8

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.084 in (2.1 mm)	1.297 in (32.94 mm)	Yes

16123-2.05.D.7.b

Overall Cable Diameter (Nominal):	1.297 in (32.94 mm)
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16123-2.05.D.7.a

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR
Pair(s)	3.67 Ohm/1000ft (12.0 Ohm/km)

Nom Outer Shield DCR@Cond DCR1-cnvr:	2.95 Ohm/1000ft (9.68 Ohm/km)
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Voltage

UL Voltage Rating
600 V (TC-ER) 150 V (NPLF)

16123-2.05.D.3

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 75°C Wet	-30C to 90C Dry, 75C wet

Bend Radius

16123-2.05.D.4

Stationary Min.	Installation Min.
15.6 in (396 mm)	15.6 in (396 mm)

Max. Pull Tension:	2606 lbs (1182 kg)
Bulk Cable Weight:	1020 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL1685 (FT4 Loading), FT4, 1202
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER16123-2.05.D.5.a (7.a)
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.559 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1089A 0101250	Black	Reel	1,250 ft	612825106203	C Z
1089A 0102500	Black	Reel	2,500 ft	612825106210	
1089A 0105000	Black	Reel	5,000 ft	612825106234	

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1055A](#) 

Inst, 2 Pr #16 Str BC, PVC-NYL Ins E1, IS/OS, Blk PVC Jkt, 600V TC-ER
150V NPLF 90C Dry/Wet

 [Request Sample](#)

Product Description

UL Instrumentation, 2 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Individual & Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER 150V NPLF 90C Dry/Wet SUN RES DIR BUR

Technical Specifications

Product Overview

Suitable Applications: Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners

Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	2	16 AWG	7x24	BC - Bare Copper

16123-2.05.D.5.b

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	ICEA Chart E1

16123-2.05.D.5.c.1

Inner Shield

Element	Shield Type	Material	Coverage	Drainwire Type	Notes
Pair(s)	Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC	each pair, extra Polyester tape for isolation.

16123-2.05.D.6.a

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

16123-2.05.D.8

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.047 in (1.2 mm)	0.475 in (12.1 mm)	Yes

16123-2.05.D.7.b

Overall Cable Diameter (Nominal): 0.475 in (12.1 mm)

16123-2.05.D.7.a

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Inner Shield DCR	Max. Current
Pair(s)	4.2 Ohm/1000ft	5.49 Ohm/1000ft (18.0 Ohm/km)	14 Amps per conductor @ 30°C

Nom Outer Shield DCR@Cond DCR1-cnvt: 3.24 Ohm/1000ft (10.6 Ohm/km)

Voltage

UL Voltage Rating
600 V (TC-ER), 150 V (NPLF)

16123-2.05.D.3

Mechanical Characteristics

Temperature

16123-2.05.D.4	
UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to 90°C Dry/Wet

Bend Radius

Stationary Min.	Installation Min.
5.7 in (140 mm)	5.7 in (140 mm)

Max. Pull Tension:	200 lbs (91 kg)
Bulk Cable Weight:	101 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a	
Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 FT4 Loading, FT4, IEEE 1202 FT4
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER16123-2.05.D.5.a (7.a)
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.629 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1055A 0107500	Black	Reel	7,500 ft	612825105510	C Z

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1039A](#) 

Inst, 4 Pr #16 Str BC, PVC-NYL Ins E1, IS/OS, Blk PVC Jkt, 600V TC-ER
150V NPLF 90C Dry/Wet

 [Request Sample](#)

Product Description

UL Instrumentation, 4 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Individual & Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER, 150V NPLF 90C Dry/Wet, SUN RES, DIR BUR

Technical Specifications

Product Overview

Suitable Applications: Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners

Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	4	16 AWG	7x24	BC - Bare Copper

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	ICEA Chart E1

Inner Shield

Element	Shield Type	Material	Coverage	Drainwire Type	Notes
Pair(s)	Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC	each pair

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.064 in (1.6 mm)	0.574 in (14.6 mm)	Yes

Overall Cable Diameter (Nominal): 0.574 in (14.6 mm)

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Inner Shield DCR	Max. Current
Pair(s)	4.2 Ohm/1000ft	5.49 Ohm/1000ft (18.0 Ohm/km)	13 Amps per conductor @ 30°C

Nom Outer Shield DCR@Cond DCR1-cnvt: 3.14 Ohm/1000ft (10.3 Ohm/km)

Voltage

UL Voltage Rating
600 V (TC-ER), 150 V (NPLF)

Mechanical Characteristics

Temperature

16123-2.05.D.4	
UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to 90°C Dry/Wet

Bend Radius

Stationary Min.	Installation Min.
6.9 in (180 mm)	6.9 in (180 mm)

Max. Pull Tension:	344 lbs (156 kg)
Bulk Cable Weight:	188 lbs/1000ft

Standards and Compliance

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 FT4 Loading, FT4, IEEE 1202 FT4
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)
Other Standard Compliance(s):	07-KA080011-MSHA

History

Update and Revision:	Revision Number: 0.620 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1039A 0107500	Black	Reel	7,500 ft	612825105206	C Z

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1040A](#) 

Inst, 6 Pr #16 Str BC, PVC-NYL Ins E1, IS/OS, Blk PVC Jkt, 600V TC-ER
150V NPLF 90C Dry/Wet

 [Request Sample](#)

Product Description

UL Instrumentation, 6 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Individual & Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER, 150V NPLF 90C Dry/Wet, SUN RES, DIR BUR

Technical Specifications

Product Overview

Suitable Applications: Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners

Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	6	16 AWG	7x24	BC - Bare Copper

16123-2.05.D.5.b

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	ICEA Chart E1

16123-2.05.D.5.c.1

Inner Shield

Element	Shield Type	Material	Coverage	Drainwire Type	Notes
Pair(s)	Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC	each pair

16123-2.05.D.6.a

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

16123-2.05.D.8

16123-2.05.D.8

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.064 in (1.6 mm)	0.670 in (17.0 mm)	Yes

16123-2.05.D.7.b

Overall Cable Diameter (Nominal): 0.670 in (17.0 mm)

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Inner Shield DCR	Nom. Inductance	Max. Current
Pair(s)	4.2 Ohm/1000ft	5.49 Ohm/1000ft (18.0 Ohm/km)	0.182 µH/ft (0.597 µH/m)	9 Amps per conductor @ 30°C

Nom Outer Shield DCR@Cond DCR1-cnvt: 3.01 Ohm/1000ft (9.88 Ohm/km)

Voltage

UL Voltage Rating
600 V (TC-ER), 150 V (NPLF)

16123-2.05.D.3

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to 90°C Dry/Wet

Bend Radius16123-2.05.D.4

Stationary Min.	Installation Min.
8.0 in (200 mm)	8.0 in (200 mm)

Max. Pull Tension:	486 lbs (220 kg)
Bulk Cable Weight:	264 lbs/1000ft

Standards and Compliance16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 FT4 Loading, FT4, IEEE 1202 FT4
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER16123-2.05.D.7.a
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)
Other Standard Compliance(s):	07-KA080011-MSHA

History

Update and Revision:	Revision Number: 0.618 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1040A 0105000	Black	Reel	5,000 ft	612825105213	C Z

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1042A](#) 

Inst, 12 Pr #16 Str BC, PVC-NYL Ins E1, IS/OS, Blk PVC Jkt, 600V TC-ER
150V NPLF 90C Dry/Wet

 [Request Sample](#)

Product Description

UL Instrumentation, 12 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Individual & Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER 150V NPLF 90C Dry/75C Wet SUN RES DIR BUR

Technical Specifications

Product Overview

Suitable Applications: Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners

Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	12	16 AWG	7x24	BC - Bare Copper

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	ICEA Chart E1

Inner Shield

Element	Shield Type	Material	Coverage	Drainwire Type	Notes
Pair(s)	Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC	each pair

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.084 in (2.1 mm)	0.907 in (23.0 mm)	Yes

Overall Cable Diameter (Nominal): 0.907 in (23.0 mm)

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Inner Shield DCR	Max. Current
Pair(s)	4.2 Ohm/1000ft	5.49 Ohm/1000ft (18.0 Ohm/km)	8 Amps per conductor @ 30°C

Nom Outer Shield DCR@Cond DCR1-cnvt: 3.14 Ohm/1000ft (10.3 Ohm/km)

Voltage

UL Voltage Rating
600 V (TC-ER), 150 V (NPLF)

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to 90°C Dry, 90°C wet

Bend Radius16123-2.05.D.4

Stationary Min.	Installation Min.
10.9 in (277 mm)	10.9 in (277 mm)

Max. Pull Tension:	915 lbs (415 kg)
Bulk Cable Weight:	491 lbs/1000ft

Standards and Compliance16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 FT4 Loading, FT4, IEEE 1202 FT416123-2.05.D.7.a
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.638 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1042A 0102500	Black	Reel	2,500 ft	612825105251	C Z
1042A 0105000	Black	Reel	5,000 ft	612825105268	C Z

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1045A](#) 

Inst, 24 Pr #16 Str BC, PVC-NYL Ins E1, IS/OS, Blk PVC Jkt, 600V TC-ER
150V NPLF 90C Dry/Wet

 [Request Sample](#)

Product Description

UL Instrumentation, 24 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Individual & Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER, 150V NPLF 90C Dry/Wet, SUN RES, DIR BUR

Technical Specifications

Product Overview

Suitable Applications: Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners

Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	24	16 AWG	7x24	BC - Bare Copper

16123-2.05.D.5.b

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	ICEA Chart E1

16123-2.05.D.5.c.1

Inner Shield

Element	Shield Type	Material	Coverage	Drainwire Type	Notes
Pair(s)	Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC	each pair

16123-2.05.D.6.a

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

16123-2.05.D.8

16123-2.05.D.8

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.084 in (2.1 mm)	1.244 in (31.60 mm)	Yes

16123-2.05.D.7.b

Overall Cable Diameter (Nominal): 1.244 in (31.60 mm)

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Inner Shield DCR	Max. Current
Pair(s)	4.2 Ohm/1000ft	5.49 Ohm/1000ft (18.0 Ohm/km)	6 Amps per conductor @ 30°C

Nom Outer Shield DCR@Cond DCR1-cnvt: 3.14 Ohm/1000ft (10.3 Ohm/km)

Voltage

UL Voltage Rating
600 V (TC-ER), 150 V (NPLF)

16123-2.05.D.3

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to 90°C Dry/Wet

Bend Radius16123-2.05.D.4

Stationary Min.	Installation Min.
14.9 in (378 mm)	14.9 in (378 mm)

Max. Pull Tension:	1772 lbs (803.8 kg)
Bulk Cable Weight:	899 lbs/1000ft

Standards and Compliance

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 FT4 Loading, FT4, IEEE 1202 FT4
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER16123-2.05.D.7.a
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.613 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1045A 0101250	Black	Reel	1,250 ft	612825105312	C Z
1045A 0105000	Black	Reel	5,000 ft	612825105329	

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1046A](#)

Inst, 36 Pr #16 Str BC, PVC-NYL Ins E1, IS/OS, Blk PVC Jkt, 600V TC-ER
150V NPLF TFN 90C Dry / 75C Wet

Request Sample

Product Description

UL Instrumentation, 36 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Individual & Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER, 150V NPLF TFN 90C Dry 75C Wet, SUN RES, DIR BUR

Technical Specifications

Product Overview

Suitable Applications:	Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners
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Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	36	16 AWG	7x24	BC - Bare Copper

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	ICEA Chart E1

Inner Shield

Element	Shield Type	Material	Coverage	Drainwire Type
Pair(s)	Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.083 in (2.1 mm)	1.471 in (37.36 mm)	Yes

Overall Cable Diameter (Nominal):	1.471 in (37.36 mm)
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Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Inner Shield DCR	Max. Current
Pair(s)	4.2 Ohm/1000ft	5.49 Ohm/1000ft (18.0 Ohm/km)	6 Amps per conductor @ 30°C

Voltage

UL Voltage Rating
600 V (TC-ER), 150 V (NPLF)

Mechanical Characteristics

Temperature

UL Temperature	Operating
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90°C Dry, 75°C Wet, -30°C to 90°C Dry, 90° wet

Bend Radius

16123-2.05.D.4

Stationary Min.	Installation Min.
17.4 in (442 mm)	17.4 in (442 mm)

Max. Pull Tension:	2629 lbs (1192 kg)
Bulk Cable Weight:	1.274 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685, FT4 Loading, FT4, IEEE 1202 FT4
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.586 Revision Date: 01-04-2024
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Product: [3090AE](#) 

Inst, 1 Pr #16 Str BC +GRN GND, PVC-NYL Ins E1, OS, Blk PVC Jkt, 600V TC-ER 150V NPLF 90C Dry

 [Request Sample](#)

Product Description

UL Instrumentation TC-ER, 1 Pair 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code + Green Ground, Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER 150V NPLF 90C Dry 90C Wet SUN RES DIR BUR

Technical Specifications

Product Overview

Suitable Applications: Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), encoders, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, Actuators, Positioners

Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Pair(s)	1	16	7x24	BC - Bare Copper
Ground	1	16	7x24	BC - Bare Copper

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Pair(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	Black & White
Conductor(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	Green

Inner Shield

Element	Shield Type	Material	Coverage	Drainwire Type
Each Pair	Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.048 in (1.2 mm)	0.375 in (9.53 mm)	Yes

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Inner Shield DCR	Nom. Capacitance Cond-to-Cond	Nom. Capacitance Cond-to-Other (Conds + Shield)	Nom. Inductance	Max. Current
Pair(s)	4.26 Ohm/1000ft	0.84 Ohm/1000ft (2.8 Ohm/km)	87 pF/ft (290 pF/m)	50 pF/ft (160 pF/m)	0.176 µH/ft (0.577 µH/m)	18 Amps per Conductor at 30°C
Ground	4.26 Ohm/1000ft (14.0 Ohm/km)					18 Amps per Conductor at 30°C

Voltage

UL Voltage Rating
600 V (TC), 150 V (NPLF)

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to 90°C Dry/Wet

Bend Radius

Stationary Min.	Installation Min.
1.48 in (37.6 mm)	1.85 in (47.0 mm)

Max. Pull Tension:	130 lbs (59 kg)
Bulk Cable Weight:	75 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 UL loading, T-29-520, IEEE 1202 FT4
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

Product Notes

Notes:	Alternate color coding available upon request.
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History

Update and Revision:	Revision Number: 0.304 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC
3090AE 0101000	Black	Reel	1,000 ft	612825141228
3090AE 0105000	Black	Reel	5,000 ft	612825141242

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Product: [3091A](#) 

Inst, 1 Tri #16 Str BC, PVC-NYL Ins E1, OS, Blk PVC Jkt, 600V TC-ER 150V NPLF 90C Dry/Wet

 [Request Sample](#)

16123-2.05.D.1

Product Description

UL Instrumentation, 1 Triad 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER 150V NPLF 90C Dry/Wet [SUN RES](#) DIR BUR

Technical Specifications

16123-2.05.D.7.a

Product Overview

Suitable Applications:	Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), RTD, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, actuators, positioners
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Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Triad(s)	1	16 AWG	7x24	BC - Bare Copper

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Triad(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	Black, White, Red

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.047 in (1.2 mm)	0.315 in (8.00 mm)	Yes

Overall Cable Diameter (Nominal):	0.315 in (8.00 mm)
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Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Max. Current
Triad(s)	4.45 Ohm/1000ft	6.2 Amps per conductor at 25°C

Nom Outer Shield DCR@Cond DCR1-cnvr:	5.47 Ohm/1000ft (17.9 Ohm/km)
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Voltage

UL Voltage Rating
600 V (TC-ER) 150 V (NPLF)

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to +90°C Dry, 75°C wet

Bend Radius

16123-2.05.D.4

Stationary Min.	Installation Min.
3.8 in (97 mm)	3.8 in (97 mm)

Max. Pull Tension:	129 lbs (58.5 kg)
Bulk Cable Weight:	62 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 UL Loading, IEEE 1202, FT4
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER16123-2.05.D.7.a
European Directive Compliance:	EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.601 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC
3091A 01010000	Black	Reel	10,000 ft	612825141273

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Product: [1099A](#) 

Inst, 12 Tri #16 Str BC, PVC-NYL Ins E1, IS/OS, Blk PVC Jkt, 600V TC-ER
150V NPLF 90C Dry/Wet

 [Request Sample](#)

Product Description

UL Instrumentation, 12 Triad 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Individual & Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER 150V NPLF 90C Dry/Wet SUN RES DIR BUR

Technical Specifications

Product Overview

Suitable Applications:	Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), RTD, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, actuators, positioners
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Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Triad(s)	12	16 AWG	7x24	BC - Bare Copper

Insulation

Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Triad(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	ICEA Chart E1

Inner Shield

Element	Shield Type	Material	Coverage	Drainwire Type	Notes
Triad(s)	Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC	each triad

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.089 in (2.3 mm)	1.065 in (27.05 mm)	Yes

Overall Cable Diameter (Nominal):	1.065 in (27.05 mm)
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Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Inner Shield DCR	Max. Current
Triad(s)	4.36 Ohm/1000ft	4.99 Ohm/1000ft (16.4 Ohm/km)	7 Amps per conductor @ 30°C

Nom Outer Shield DCR@Cond DCR1-cnvt:	2.74 Ohm/1000ft (8.99 Ohm/km)
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Voltage

UL Voltage Rating
600 V (TC-ER) 150 V (NPLF)

Mechanical Characteristics

Temperature

UL Temperature	Operating
90°C Dry, 90°C Wet	-30°C to 90°C Dry/Wet

Bend Radius16123-2.05.D.4

Stationary Min.	Installation Min.
12.8 in (325 mm)	12.8 in (325 mm)

Max. Pull Tension:	1343 lbs (609.2 kg)
Bulk Cable Weight:	674 lbs/1000ft

Standards and Compliance16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL 1685 FT4 Loading, FT4, IEEE 1202 FT416123-2.05.D.7.a
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC-ER
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.627 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1099A 0102000	Black	Reel	2,000 ft	612825106753	C Z
1099A 0102500	Black	Reel	2,500 ft	612825106760	C Z

Footnote:	C - CRATE REEL PUT-UP., Z - FINAL PUT-UP MAY VARY (= OR -) 10% FOR SPOOLS OR REELS AND (+ OR -) 5% FOR UNREEL CARTONS FROM LENGTH SHOWN.
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Product: [1100A](#)

Inst, 24 Tri #16 Str BC, PVC-NYL Ins E1, IS/OS, Blk PVC Jkt, 600V TC 150V NPLF 90C Dry 75C Wet

[Request Sample](#)

Product Description

UL Instrumentation, 24 Triad 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Individual & Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC 150V NPLF 90C Dry 75C Wet SUN RES DIR BUR

Technical Specifications

Product Overview

Suitable Applications:	Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...),low voltage digital control (24v, ...),RTD, control circuits, distributed control system (DCS),programmable logic controller (PLC), Solenoids, Valves, actuators, positioners
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Construction Details

Conductor				
Element	No. of Elements	Size	Stranding	Material
Triad(s)	24	16 AWG	7x24	BC - Bare Copper
Insulation				
Element	Material	Nom. Thickness	Nom. Insulation Diameter	Color Code
Triad(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	0.101 in (2.57 mm)	Black, White, Red and Numbered
Inner Shield				
Element	Shield Type	Material	Coverage	Drainwire Type
Triad(s)	Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC
Outer Shield				
Shield Type	Material	Coverage	Drainwire Type	
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC	
Outer Jacket				
Material	Nom. Thickness	Nom. Diameter	Ripcord	
PVC - Polyvinyl Chloride	0.084 in (2.1 mm)	1.447 in (36.75 mm)	Yes	
Overall Cable Diameter (Nominal): 1.447 in (36.75 mm)				

Electrical Characteristics

Electricals		
Element	Nom. Conductor DCR	Nom. Inner Shield DCR
Triad(s)	3.67 Ohm/1000ft (12.0 Ohm/km)	4.99 Ohm/1000ft (16.4 Ohm/km)

Voltage	
UL Voltage Rating	
600 V (TC), 150 V (NPLF)	

Mechanical Characteristics

Temperature	
UL Temperature	Operating

90°C Dry, 75°C Wet -30C to 90C Dry, 75C wet

Bend Radius

16123-2.05.D.4

Stationary Min.	Installation Min.
14.5 in (368 mm)	14.5 in (368 mm)

Max. Pull Tension:	3144 lbs (1426 kg)
Bulk Cable Weight:	1225 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL1685 (FT4 Loading), FT4, 1202
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE)
APAC Compliance:	China RoHS II (GB/T 26572-2011)

History

Update and Revision:	Revision Number: 0.201 Revision Date: 01-04-2024
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Product: [3130A](#)

Inst, 36 Tri #16 Str BC, PVC-NYL Ins E1, IS/OS, Blk PVC Jkt, 600V TC-ER
150V NPLF 90C Dry 75C Wet

Request Sample

Product Description

UL Instrumentation, 36 Triad 16AWG (7x24) Bare Copper, PVC-NYL Insulation E1 Color Code, Individual & Overall Beldfoil® Shield, Black PVC Outer Jacket, 600V TC-ER, 150V NPLF 90C Dry 75C Wet (SUN RES) DIR BUR

Technical Specifications

Product Overview

Suitable Applications: Industrial Control, Raceways Cable trays and ducts, Digital Control, Instruments (4-20ma, 0-10v, ...), low voltage digital control (24v, ...), RTD, control circuits, distributed control system (DCS), programmable logic controller (PLC), Solenoids, Valves, actuators, positioners

Construction Details

Conductor

Element	No. of Elements	Size	Stranding	Material
Triad(s)	36	16 AWG	7x24	BC - Bare Copper

Insulation

Element	Material	Nom. Thickness	Color Code
Triad(s)	PVC/Nylon - Polyvinyl Chloride + Nylon	0.022 in (0.56 mm)	Black, White, Red and Numbered

Inner Shield

Element	Shield Type	Material	Coverage	Drainwire Type
Triad(s)	Tape	Bi-Laminate (Alum+Poly)	100%	18 AWG (7x26) TC

Outer Shield

Shield Type	Material	Coverage	Drainwire Type
Tape	Bi-Laminate (Alum+Poly)	100%	16 AWG (7x24) TC

Outer Jacket

Material	Nom. Thickness	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.110 in (2.79 mm)	1.773 in (45.03 mm)	Yes

Overall Cable Diameter (Nominal): 1.773 in (45.03 mm)

Electrical Characteristics

Electricals

Element	Nom. Conductor DCR	Nom. Inner Shield DCR
Triad(s)	3.67 Ohm/1000ft (12.0 Ohm/km)	4.99 Ohm/1000ft (16.4 Ohm/km)

Voltage

UL Voltage Rating
600 V (TC), 150 V (NPLF)

Mechanical Characteristics

Temperature

UL Temperature	Operating
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90°C Dry, 75°C Wet -30C to 90C Dry, 75C wet

Bend Radius

16123-2.05.D.4

Stationary Min.	Installation Min.
17.7 in (450 mm)	17.7 in (450 mm)

Max. Pull Tension:	3600 lbs (1600 kg)
Bulk Cable Weight:	1839 lbs/1000ft

Standards and Compliance

16123-2.05.D.7.a

Environmental Suitability:	Indoor/Outdoor, Indoor, Outdoor, Sunlight Resistance, Burial
Flammability / Reaction to Fire:	UL1685 (FT4 Loading), FT4, 1202
NEC / UL Compliance:	Article 336, Article 760, NPLF, TC
European Directive Compliance:	EU CE Mark

History

Update and Revision:	Revision Number: 0.208 Revision Date: 01-04-2024
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4.0 NETWORK CABLES

SPECIFICATION – 16123-2.03.A.4 600-VOLT OR LESS WIRES AND CABLES





Product: [3633](#) 

Category 6+ Premium Cable, 4 Bonded-Pairs, U/UTP, CMP

Product Description

Category 6+ Premium Premise Horizontal Cable (400MHz), 4 Bonded-Pairs, 23 AWG Solid Bare Copper Conductors, U/UTP, Plenum-CMP, Flamarrest® PVC-LS Jacket

Technical Specifications

Product Overview

Suitable Applications:	Premise Horizontal Cable, Ethernet 1000BASE-T, Ethernet 100BASE-TX, Ethernet 10BASE-T, PoE++, PoE+, PoE
Patent:	This product has one or more applicable patents. More information on patents can be found at https://www.belden.com/patents .

Construction Details

Conductor

Size	Stranding	Material	No. of Pairs
23 AWG	Solid	BC - Bare Copper	4

16123-2.05.E.2.f.2.a

Insulation

Material	Color Code
FEP - Fluorinated Ethylene Propylene	White/Blue Stripe & Blue, White/Orange Stripe & Orange, White/Green Stripe & Green, White/Brown Stripe & Brown

16123-2.05.E.2.f.4

Bonded-Pair: Yes

16123-2.05.E.2.f.3.b

Outer Jacket

Separator	Material	Material Trade Name	Nom. Diameter	Ripcord
Center Member (Patented X-Spline®)	PVC - Polyvinyl Chloride	Flamarrest®	0.230 in (5.84 mm)	Yes

16123-2.05.E.2.f.5

Overall Cable Diameter (Nominal): 0.230 in (5.84 mm) 16123-2.05.E.2.f.a.1

Electrical Characteristics

Electricals

Max. Conductor DCR	Max. Capacitance Unbalance
77 Ohm/km (23 Ohm/1000ft)	90 pF/100m

Delay

Frequency	Max. Delay	Max. Delay Skew	Nom. Velocity of Prop.	Typical Delay Skew
100 MHz	537.6 ns/100m	35 ns/100m	72%	30 ns/100m

High Frequency

Frequency [MHz]	Max. Insertion Loss (Attenuation)	Min. NEXT [dB]	Min. PSNEXT [dB]	Min. ACR [dB]	Min. PSACR [dB]	Min. ACRF (ELFEXT) [dB]	Min. PSACRF (PSELFEXT) [dB]	Min. RL (Return Loss) [dB]	Max./Min. Input Impedance (unFitted) [Ohm]	Max./Min. Fitted Impedance [Ohm]
1	2.0 dB/100m	84.3	82.3	80.3	80.3	76.8	73.8	20.0	100 ± 15	
4	3.7 dB/100m	75.3	73.3	69.5	69.5	64.8	61.8	23.6	100 ± 15	100 ± 12
8	5.3 dB/100m	70.8	68.8	63.5	63.5	58.7	55.7	25.3	100 ± 15	100 ± 10.4
10	5.9 dB/100m	69.3	67.3	61.4	61.4	56.8	53.8	26.0	100 ± 15	100 ± 8
16	7.4 dB/100m	66.2	64.2	56.8	56.8	52.7	49.7	26.0	100 ± 15	100 ± 7.3
20	8.3 dB/100m	64.8	62.8	54.4	54.4	50.8	47.8	26.0	100 ± 15	100 ± 5.7
25	9.4 dB/100m	63.3	61.3	52.0	52.0	48.8	45.8	25.5	100 ± 15	100 ± 5

31.25	10.5 dB/100m	61.9	59.9	49.4	49.4	46.9	43.9	25.0	100 ± 15	100 ± 5
62.5	15.1 dB/100m	57.4	55.4	40.3	40.3	40.9	37.9	23.5	100 ± 15	100 ± 5
100	19.3 dB/100m	54.3	52.3	33.0	33.0	36.8	33.8	22.5	100 ± 15	100 ± 5
155	24.5 dB/100m	51.4	49.4	24.9	24.9	33.0	30.0	21.6	100 ± 22	100 ± 5
200	28.2 dB/100m	49.8	47.8	19.6	19.6	30.8	27.8	21.0	100 ± 22	100 ± 5
250	31.8 dB/100m	48.3	46.3	14.5	14.5	28.8	25.8	20.5	100 ± 32	100 ± 5
300	35.2 dB/100m	47.1	45.1	9.9	9.9	27.3	24.3	20.1	100 ± 32	100 ± 5
350	38.4 dB/100m	46.1	44.1	5.7	5.7	25.9	22.9	19.8	100 ± 32	100 ± 5
400	41.5 dB/100m	45.3	43.3	1.8	1.8	24.8	21.8	19.5	100 ± 32	100 ± 5
450	44.3 dB/100m	44.5	42.5			23.7	20.7	19.2	100 ± 32	100 ± 5
500	47.1 dB/100m	43.8	41.8			22.8	19.8	19.0	100 ± 32	100 ± 5
550	49.7 dB/100m	43.2	41.2			22.0	19.0	18.8	100 ± 32	100 ± 5
600	52.3 dB/100m	42.6	40.6			21.2	18.2	18.7	100 ± 32	100 ± 5
650	54.8 dB/100m	42.1	40.1			20.5	17.5	18.5	100 ± 32	100 ± 5

Voltage

UL Voltage Rating
300 V (CMP), 300 V (CL3P)
16123-2.05.E.2.f.7.a

Mechanical Characteristics

Temperature

UL Temperature	Operating	Installation	Storage
105°C	-20°C To +75°C	0°C To +50°C	-20°C To +75°C

Bend Radius

Stationary Min.	Installation Min.
0.5 in (13 mm)	2.5 in (64 mm)

Max. Pull Tension:	40 lbs (18 kg)
Bulk Cable Weight:	31 lbs/1000ft

Standards and Compliance

Environmental Suitability:	Plenum Indoor	2.05.E.2.f.3.b
Sustainability:	Product Lens™, Environmental Product Declaration (EPD) Available	
Flammability / Reaction to Fire:	NFPA 262, UL 910 (Plenum), FT6, IEC 60332-1-2	
CPR Compliance:	CPR Euroclass: Eca; CPR UKCA Class: Eca	16123-2.05.E.2.f.8.c
NEC / UL Compliance:	Article 800, CMP, CMP-LP (0.6A), CL3P-LP (0.6A)	
CEC / C(UL) Compliance:	CMP	16123-2.05.E.2.f.8.c
ICEA Compliance:	S-116-732	16123-2.05.E.2.f.9
IEEE Compliance:	IEEE 802.3bt Type 1, Type 2, Type 3, Type 4	
NEMA Compliance:	ANSI/NEMA WC-86	
Data Category:	Category 6	16123-2.05.E.2.f.9
TIA/EIA Compliance:	ANSI/TIA-568.2-D Category 6	16123-2.05.E.2.f.1
Third Party Performance Verification:	Category 6	
ISO/IEC Compliance:	ISO/IEC 11801-1, IEC 61156-5	
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), REACH, EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE), REACH: 2020-01-16	
UK Regulation Compliance:	UKCA Mark	
APAC Compliance:	China RoHS II (GB/T 26572-2011)	
Non-Plenum Number:	3632	

Product Notes

Notes:	Electrical values are expected performance based on cable testing and representative performance within a typical Belden system. Values above 400 MHz are for Engineering Information Only. Print Includes Descending Footage/Meter Markings from Max. Put-Up Length to 0.
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History

Update and Revision:	Revision Number: 0.538 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
3633 0101000	Black	Reel	1,000 ft	612825146421	C
3633 010A1000	Black	Reel-in-Box	1,000 ft	612825146414	C
3633 D151000	Blue	Reel	1,000 ft	612825146261	C
3633 D15A1000	Blue	Reel-in-Box	1,000 ft	612825146247	C
3633 D15U1000	Blue	UnReel	1,000 ft	612825146254	
3633 0081000	Gray	Reel	1,000 ft	612825146377	C
3633 008A1000	Gray	Reel-in-Box	1,000 ft	612825146353	C
3633 008U1000	Gray	UnReel	1,000 ft	612825146360	
3633 0051000	Green	Reel	1,000 ft	612825146346	C
3633 005A1000	Green	Reel-in-Box	1,000 ft	612825146339	C
3633 0031000	Orange	Reel	1,000 ft	612825146300	C
3633 003A1000	Orange	Reel-in-Box	1,000 ft	612825146292	C
3633 0021000	Red	Reel	1,000 ft	612825146285	C
3633 002A1000	Red	Reel-in-Box	1,000 ft	612825146278	C
3633 0091000	White	Reel	1,000 ft	612825146407	C
3633 009A1000	White	Reel-in-Box	1,000 ft	612825146384	C
3633 009U1000	White	UnReel	1,000 ft	612825146391	
3633 0041000	Yellow	Reel	1,000 ft	612825146322	C
3633 004A1000	Yellow	Reel-in-Box	1,000 ft	612825146315	C

Footnote: C - CRATE REEL PUT-UP.

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Product: [1872A](#) 

Category 6+ Cable, 4 Bonded-Pairs, U/UTP, CMR

Product Description

Category 6+ Premise Horizontal Cable (350MHz), 4 Bonded-Pairs, 23 AWG Solid Bare Copper Conductors, U/UTP, Riser-CMR, PVC Jacket

Technical Specifications

Product Overview

Suitable Applications:	Premise Horizontal Cable, Ethernet 1000BASE-T, Ethernet 100BASE-TX, Ethernet 10BASE-T, PoE++, PoE+, PoE
Patent:	This product has one or more applicable patents. More information on patents can be found at https://www.belden.com/patents .

Construction Details

Conductor

Size	Stranding	Material	No. of Pairs
23 AWG	Solid	BC - Bare Copper	4

16123-2.05.E.2.f.2.a

Insulation

Material	Color Code
PO - Polyolefin	White/Blue Stripe & Blue, White/Orange Stripe & Orange, White/Green Stripe & Green, White/Brown Stripe & Brown

16123-2.05.E.2.f.4

Bonded-Pair:	Yes
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Outer Jacket

16123-2.05.E.2.f.3.a

Material	Nom. Diameter	Ripcord
PVC - Polyvinyl Chloride	0.365 x 0.165 in	Yes

16123-2.05.E.2.f.5

Overall Cable Diameter (Nominal):	0.365 x 0.165 in
-----------------------------------	------------------

16123-2.05.E.2.f.a.1

Electrical Characteristics

Electricals

Max. Conductor DCR	Max. Capacitance Unbalance
75 Ohm/km (23 Ohm/1000ft)	49 pF/100m

Delay

Frequency	Max. Delay	Max. Delay Skew	Nom. Velocity of Prop.
100 MHz	537.6 ns/100m	25 ns/100m	70%

High Frequency

Frequency [MHz]	Max. Insertion Loss (Attenuation)	Min. NEXT [dB]	Min. PSNEXT [dB]	Min. ACR [dB]	Min. PSACR [dB]	Min. ACRF (ELFEXT) [dB]	Min. PSACRF (PSELFEXT) [dB]	Min. RL (Return Loss) [dB]	Max./Min. Input Impedance (unFitted) [Ohm]	Max./Min. Fitted Impedance [Ohm]
1	1.9 dB/100m	75.3	75.3	73.4	73.4	67.8	64.8	20.0	100 ± 12	100 ± 15
4	3.7 dB/100m	66.3	66.3	62.6	62.6	55.7	52.8	23.0	100 ± 12	100 ± 15
8	5.3 dB/100m	61.8	61.8	56.5	56.5	49.7	46.7	24.5	100 ± 12	100 ± 15
10	5.9 dB/100m	60.3	60.3	54.4	54.4	47.8	44.8	25.0	100 ± 12	100 ± 15
16	7.5 dB/100m	57.2	57.2	49.7	49.7	43.7	40.7	25.0	100 ± 12	100 ± 8
20	8.4 dB/100m	55.8	55.8	47.3	47.3	41.8	38.8	25.0	100 ± 12	100 ± 8
25	9.5 dB/100m	54.3	54.3	44.9	44.9	39.8	36.8	24.3	100 ± 15	100 ± 8
31.25	10.6 dB/100m	52.9	52.9	42.2	42.2	37.9	34.9	23.6	100 ± 15	100 ± 8

62.5	15.4 dB/100m	48.4	48.4	33.0	33.0	31.8	28.9	21.5	100 ± 15	100 ± 8
100	19.8 dB/100m	45.3	45.3	25.5	25.5	27.8	24.8	21.0	100 ± 15	100 ± 8
155	25.1 dB/100m	42.4	42.4	17.3	17.3	24.0	21.0	21.0	100 ± 15	100 ± 8
200	29.0 dB/100m	40.8	40.8	11.8	11.8	21.7	18.8	21.0	100 ± 15	100 ± 8
250	32.8 dB/100m	39.3	39.3	6.5	6.5	19.8	16.8	18.0	100 ± 20	100 ± 8
300	36.4 dB/100m	38.1	38.1	1.7	1.7	18.3	15.3	18.0	100 ± 20	100 ± 8
310	39.8 dB/100m	37.1	37.1	-2.6	-2.6	16.9	13.9	17.0	100 ± 22	100 ± 8
350	43.0 dB/100m	36.3	36.3			15.8	12.8	14.0	100 ± 32	100 ± 8
400	46.0 dB/100m	35.5	35.5			14.7	11.7	14.0	100 ± 32	100 ± 8
500	48.9 dB/100m	34.8	34.8			13.8	10.8	14.0	100 ± 32	100 ± 8
550	51.8 dB/100m	34.2	34.2			13.0	10.0	14.0	100 ± 32	100 ± 8

Voltage

UL Voltage Rating	
300 V (CMR)	16123-2.05.E.2.f.7.a

Mechanical Characteristics

Temperature

UL Temperature	Operating	Installation	Storage
75°C	-20°C To +75°C	0°C To +50°C	-20°C To +75°C

Bend Radius

Stationary Min.	Installation Min.
0.5 in (13 mm)	1.75 in (44.5 mm)

Max. Pull Tension:	40 lbs (18 kg)
Bulk Cable Weight:	32 lbs/1000ft

Standards and Compliance

Environmental Suitability:	Riser, Indoor
Sustainability:	Product Lens™, Environmental Product Declaration (EPD) Available
Flammability / Reaction to Fire:	UL 1666 Riser, FT4, IEC 60332-1-2
CPR Compliance:	CPR Euroclass: Eca; CPR UKCA Class: Eca
NEC / UL Compliance:	Article 800, CMR
CEC / C(UL) Compliance:	CMR
ICEA Compliance:	S-116-732
IEEE Compliance:	IEEE 802.3bt Type 1, Type 2, Type 3, Type 4
NEMA Compliance:	ANSI/NEMA WC-66
Data Category:	Category 6
TIA/EIA Compliance:	ANSI/TIA-568.2-D Category 6
Third Party Performance Verification:	Category 6
ISO/IEC Compliance:	ISO/IEC 11801-1, IEC 61156-5
European Directive Compliance:	EU CE Mark, EU Directive 2015/863/EU (RoHS 2 amendment), REACH, EU Directive 2011/65/EU (RoHS 2), EU Directive 2012/19/EU (WEEE), REACH: 2020-01-16
UK Regulation Compliance:	UKCA Mark
APAC Compliance:	China RoHS II (GB/T 26572-2011)
Plenum Number:	1874A

Product Notes

Notes:	Electrical values are expected performance based on cable testing and representative performance within a typical Belden system. Values above 350 MHz are for Engineering Information Only. Print Includes Descending Footage Markings from Max. Put-Up Length to 0.
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History

Update and Revision:	Revision Number: 0.539 Revision Date: 09-07-2025
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Part Numbers

Variants

Item #	Color	Putup Type	Length	UPC	Footnote
1872A 0101000	Black	Reel	1,000 ft	612825125464	C

1872A 0061000	Blue	Reel	1,000 ft	612825125419	C
1872A 006A1000	Blue	Reel-in-Box	1,000 ft	612825125402	
1872A X06G1000	Gold	Reel	1,000 ft	612825125310	C
1872A X06GA1000	Gold	Reel-in-Box	1,000 ft	612825125303	
1872A F01H1000	Gray	Reel	1,000 ft	612825125297	C
1872A F01HA1000	Gray	Reel-in-Box	1,000 ft	612825320647	
1872A 0051000	Green	Reel	1,000 ft	612825125396	C
1872A 005A1000	Green	Reel-in-Box	1,000 ft	612825125389	
1872A 0031000	Orange	Reel	1,000 ft	612825125358	C
1872A 003A1000	Orange	Reel-in-Box	1,000 ft	612825125341	
1872A 0071000	Purple	Reel	1,000 ft	612825125433	C
1872A 007A1000	Purple	Reel-in-Box	1,000 ft	612825125426	
1872A 0021000	Red	Reel	1,000 ft	612825125334	C
1872A 002A1000	Red	Reel-in-Box	1,000 ft	612825125327	
1872A 0091000	White	Reel	1,000 ft	612825125457	C
1872A 009A1000	White	Reel-in-Box	1,000 ft	612825125440	
1872A 0041000	Yellow	Reel	1,000 ft	612825125372	C
1872A 004A1000	Yellow	Reel-in-Box	1,000 ft	612825125365	

Footnote: C - CRATE REEL PUT-UP.

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Detailed Specifications & Technical Data

ENGLISH MEASUREMENT VERSION



3107A Multi-Conductor - EIA Industrial RS-485 PLTC/CM

16123-2.05.E.3

For more Information
please call

1-800-Belden1



16123-2.05.E.3.a.3

Description:

22 AWG stranded (7x30) tinned copper conductors, Datalene® insulation, twisted pairs, overall Beldfoil® shield (100% coverage) plus a tinned copper braid (65% coverage), drain wire, UV resistant PVC jacket.

16123-2.05.E.3.a.1

Physical Characteristics (Overall)

Conductor

AWG:

# Pairs	AWG	Stranding	Conductor Material	Dia. (in.)
2	22	7x30	TC - Tinned Copper	.030

Insulation

Insulation Material:

16123-2.05.E.3.a.2

16123-2.05.E.3.a

Insulation Trade Name	Insulation Material
Datalene®	FHDPE - Foam High Density Polyethylene

16123-2.05.E.3.a.4.a

Outer Shield

Outer Shield Material:

Layer #	Outer Shield Trade Name	Type	Outer Shield Material	Coverage (%)
1	Beldfoil®	Tape	Aluminum Foil-Polyester Tape	100.000
2		Braid	TC - Tinned Copper	65.000

16123-2.05.E.3.a.5

Outer Shield Drain Wire AWG:

AWG	Stranding	Drain Wire Conductor Material
22	7x30	TC - Tinned Copper

16123-2.05.E.3.a.6

Outer Shield Drain Wire Diameter:

.030

Outer Jacket

Outer Jacket Material:

Outer Jacket Material
PVC - Polyvinyl Chloride

16123-2.05.E.3.a.7

Overall Cable

Overall Cabling Color Code Chart:

Number	Color
1	White/Blue Stripe & Blue/White Stripe
2	White/Orange Stripe & Orange/White Stripe

Overall Nominal Diameter:

0.350 in.

Mechanical Characteristics (Overall)

Operating Temperature Range: -20°C To +60°C

Non-UL Temperature Rating: 60°C

Bulk Cable Weight: 73 lbs/1000 ft.

Max. Recommended Pulling Tension: 80 lbs.

Min. Bend Radius (Install)/Minor Axis: 3.500 in.

16123-2.05.E.3.a.9

Applicable Specifications and Agency Compliance (Overall)

Applicable Standards & Environmental Programs

NEC/(UL) Specification:

CM, PLTC OIL RES II

3107A Multi-Conductor - EIA Industrial RS-485 PLTC/CM

CEC/C(UL) Specification:	CM
EU CE Mark:	Yes
EU Directive 2000/53/EC (ELV):	Yes
EU Directive 2002/95/EC (RoHS):	Yes
EU RoHS Compliance Date (mm/dd/yyyy):	10/01/2005
EU Directive 2002/96/EC (WEEE):	Yes
EU Directive 2003/11/EC (BFR):	Yes
CA Prop 65 (CJ for Wire & Cable):	Yes
MII Order #39 (China RoHS):	Yes

Flame Test

UL Flame Test:	UL1685 UL Loading	16123-2.05.E.3.a.8
CSA Flame Test:	FT1	

Suitability

Sunlight Resistance:	Yes	16123-2.05.E.3.a.7.a
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Plenum/Non-Plenum

Plenum (Y/N):	No
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Surface Printing (Overall)**Electrical Characteristics (Overall)****Nom. Characteristic Impedance:**

Impedance (Ohm)
120

Nom. Capacitance Conductor to Conductor:

Capacitance (pF/ft)
11.0

Nom. Capacitance Cond. to Other Conductor & Shield:

Capacitance (pF/ft)
20.9

Nominal Velocity of Propagation:

VP (%)
78

Nom. Conductor DC Resistance:

DCR @ 20°C (Ohm/1000 ft)
14.7

Nominal Outer Shield DC Resistance:

DCR @ 20°C (Ohm/1000 ft)
1.8

Nom. Attenuation:

Description	Freq. (MHz)	Start Freq. (MHz)	Stop Freq. (MHz)	Attenuation (dB/100 ft.)
	1			.5

Max. Operating Voltage - UL:

Voltage	
300 V RMS	16123-2.05.E.3.a.4.b

Max. Recommended Current:

Current
2.7 Amps per conductor @ 25°C

Input/Unfitted Impedance:

Description	Freq. (MHz)	Start Freq. (MHz)	Stop Freq. (MHz)	Impedance (Ohm)
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3107A Multi-Conductor - EIA Industrial RS-485 PLTC/CM

	1			
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Notes (Overall)

Notes: Oil Resistance: Passes Oil Res II Per UL1277, Table 10.17. For CPE jacketed version order YR46792.

Related Documents:

No related documents are available for this product

Put Ups and Colors:

Item #	Putup	Ship Weight	Color	Notes	Item Desc
3107A 0101000	1,000 FT	62.000 LB	BLACK	C	2 PR #22 FHDPE SH PVC
3107A 0104000	4,000 FT	256.000 LB	BLACK	C	2 PR #22 FHDPE SH PVC
3107A 0105000	5,000 FT	330.000 LB	BLACK	C N	2 PR #22 FHDPE SH PVC

Notes:

C = CRATE REEL PUT-UP.

N = FINAL PUT-UP LENGTH MAY VARY -0% TO +10% FROM LENGTH SHOWN.

Revision Number: 1 Revision Date: 03-17-2009

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Belden declares this product to be in compliance with EU LVD (Low Voltage Directive 73/23/EEC), as amended by directive 93/68/EEC.

5.0 ACCESSORIES

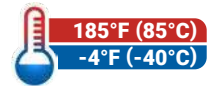
SPECIFICATION – 16123-2.06.A 600-VOLT OR LESS WIRES AND CABLES



TY-RAP® CABLE TIES

Low-profile head is designed to prevent snags on uneven surfaces and easier to pull through bulkheads. Non-magnetic stainless steel locking device insures both maximum strength and the right tightness every time.

- Because there are no notches, Ty-Rap ties provide an infinite number of locking positions while fastening the tie at the precise bundle diameter needed
- The no-slip tail provides easy grip during tensioning and the unique friction surface reduces slippage during installation
- Smooth, notchless body reduces stress concentration points under tension, making the tie stronger and lower profile than notched body ties
- Rounded edges protect insulation, preventing sharp edges from damaging cables
- Less stress in the tie body, resulting in less chance of breakage
- Engineered for labor savings and high performance
- Standard black ties are UV-Resistant nylon 6/6



Tie Color	Bulk Qty. 1000		Qty. 100		Approx. Length	Approx. Bundle Dia.		Military Standard No.	Width	Tensile Strength
	Stock No.	Thomas & Betts No.	Stock No.	Thomas & Betts No.		Min.	Max.			
Miniature										
Black	21641	TY23MX	----	----	3.6	0.06	.625	MS3367-4	.090	18 lbs.
Black	21619	TY232MX	21719	TY5232MX	8.00	0.06	2.000	----	.090	18 lbs.
Black	21617	TY234MX	----	----	14.00	0.06	4.000	----	.090	18 lbs.
Intermediate										
Black	21615	TY24MX	----	----	5.50	0.08	1.125	MS3367-5	.140	40 lbs.
Black	21693	TY242MX	----	----	8.2	0.08	2.000	----	.140	40 lbs.
Natural	21674	TY26M	----	----	11.2	0.08	3.000	----	.140	40 lbs.
Black	21694	TY26MX	----	----	11.2	0.08	3.000	----	.140	40 lbs.
Standard										
Black	21644	TY25MX	21744	TY525MX	7.3	0.12	1.750	MS3367-1	.190	50 lbs.
Black	21645	TY253MX	21745	TY5253MX	11.40	0.12	3.000	MS3367-7	.190	50 lbs.
Black	21646	TY28MX	21746	TY528MX	14.20	0.12	4.000	MS3367-2	.190	50 lbs.
* Not made in USA										
All dimensions in inches										

* Not made in USA

All dimensions in inches



Heavy-Duty										
Tie Color	Bulk Qty. 500		Qty. 50		Approx. Length	Approx. Bundle Dia.		Military Standard No.	Width	Tensile Strength
	Stock No.	Thomas & Betts No.	Stock No.	Thomas & Betts No.		Min	Max.			
Black	21690 *	TY272MX-500	----	----	8.00	0.19	2.000	----	.270	120 lbs.
Black	21691	TY27MX-500	21791	TY527MX	13.40	0.19	3.500	MS3367-3	.270	120 lbs.
Natural	21627	TY275M-500	----	----	18.00	0.19	5.000	----	.270	120 lbs.
Black	21647	TY275MX-500	21747	TY5275MX	18.00	0.19	5.000	----	.270	120 lbs.
Black	21651	TY29MX-500	21749	TY529MX	30.00	0.19	9.000	MS3367-6	.270	120 lbs.

* Not made in USA

All dimensions in inches

Cable Ties

Built for Durability:

- A wide range of sizes, styles and materials for fastening, bundling, clamping and identifying in a variety of applications
- Panduit cable ties are designed and manufactured to meet or exceed industry standards to deliver maximum reliability
- Panduit cable ties are engineered from specially selected materials for optimum reliability in the harshest environments including exposure to: acids, bases, chemicals, radiation, and all-weather conditions with operating temperatures up to 500°F (260°C)

Suitability for Securement and Support:

- Select ties meet the 2017 NEC Code for support and secureness of various cable types identified in the Code
- Select Standard cross-section ties (PLT, BT) and Light-Heavy ties (BT)
- Ties meeting this requirement are designated as UL 62275 Type 2S, 21S
 - Select Standard cross-section ties (PLT, BT) and Light-Heavy ties (BT)



High-temperature PEEK:

Provide excellent performance at continuous operating temperatures of up to 500°F (260°C), best in class strength, in addition to chemical and radiation resistance



PLT**-*71 Series

Weather Resistant Acetal:

Provides excellent ultra-violet resistance, strength and durability for long service life. Temperatures of up to 185°F (85°C)

DL**-*0 Series



Polypropylene:

Provide superior chemical resistance to acids, bases, and salts making them ideal for chemical plants. Temperatures of up to 239°F (115°C)



PLT**-*109 Series: Green, Indoor use

PLT**-*100 Series: Black, Weather Resistant

TEFZEL:

Excellent flammability rating of UL 94V-0. Distinct aqua blue color. Ideal for applications requiring resistance to gamma radiation and chemical attack. Temperatures of up to 338°F (170°C).



PLT**-*76 Series

Tefzel is a registered trademark of The Chemours Company

Installation Tools:

GTS-E

Installs subminiature, miniature, intermediate and standard cross section cable ties



GTH-E

Installs standard, heavy-standard, light-heavy, and heavy cross section cable ties





Weather Resistant Nylon 12:

A material with 12-15 years of weathering life expectancy and resistance to zinc chlorides and dilute acids. Ideal for wire management on galvanized surfaces / frames. Nylon 12 ties are marked with a "+" sign at the neck to distinguish these ties from black weather resistant nylon 6.6 ties. Temperatures of up to 194°F (90°C)



PLT**-*120 Series

HALAR:

Identifiable maroon colored parts UL listed for use in plenum or air handling spaces per NEC, Section 300-22 (C) and (D); UL62275 AH2 rating. Suitable in applications where flame resistance and smoke generation is a concern. Temperatures of up to 257°F (125°C)



PLT**-*702Y Series

HALAR is a registered trademark of Solvay Solexis

Flame Retardant Nylon 6.6:

Ties are UL94V-0 rated for applications requiring resistance to flame propagation. Temperatures of up to 212°F (100°C)

PLT**-*69 Series



Metal Barb Tie

Stainless steel lock barb provides consistent performance, reliability, and infinite adjustability through entire bundle range applications



BT**-* Series: Natural, Indoor use, up to 185°F (85°)

BT**-*0 Series: Black, Weather Resistant, up to 185°F (85°)

BT**-*30 Series: Black, Heat Stabilized up to 239°F (115°C)

Releasable Tie:

Release tab permits easy release and re-use where changes are anticipated

PRT** Series



Metal Detectable:

Unique metal impregnated material allows identification by metal detectors to meet food and beverage industry FDA requirements by helping to reduce product contamination



PLT**-*86 Series, up to 185°F (85°C)

Also available in polypropylene:

PLT**-*186 Series, up to 239°F (115°C)

*Signifies different lengths and package size options. Entering * in the part number on www.panduit.com will give you different size options.

Polywater®

J Lubricant



WATER-BASED GEL PULLING LUBRICANT



Can be used with the Polywater® LP-D5 drill-powered pump.

A Proven, High-Performance Cable Lubricant for Heavy Cable Installations

A vast amount of underground electrical and communications cable has been pulled over the past three decades using Polywater® J Lubricant. The high-performing, slow-drying, and clean nature of this water-based gel lubricant provides maximum tension reduction in all types of cable pulling, especially in hot environments. It is a proven and popular choice for long and multiple-bend cable pulls.



FEATURES AND BENEFITS

- **SUPERIOR FRICTION REDUCTION**
For low cable pulling tensions
- **HIGH CLING FACTOR**
Remains on cable jacket throughout the installation
- **FIELD-FRIENDLY APPLICATION**
Can be pumped with the LP-D5 drill-powered lubricant pump, hand applied, or deployed with Front End Packs™
- **CLEAN AND NON-STAINING**
Minimize clean-up time, especially in finished indoor use
- **UL/ULC CERTIFICATION**
UL Listed to US and Canadian safety standards

Polywater® J Lubricant



Packages for different installation environments.



CATALOG #	PACKAGE DESCRIPTION	UNITS/CASE
J-35 / WJ-35	1 qt (0.95 L) Bottle	12
J-27 / WJ-27	1 qt (0.95 L) Front End Pack™ in a Carton	12
J-99 / WJ-99	1 qt (0.95 L) Front End Pack in a Pail	16
J-55 / WJ-55	½ gal (1.9 L) Front End Pack in a Carton	6
J-110 / WJ-110	½ gal (1.9 L) Front End Pack in a Pail	10
J-128 / WJ-128	1 gal (3.8 L) Pail	4
J-640 / WJ-640	5 gal (18.9 L) Pail	1
J-DRUM / WJ-DRUM	55 gal (208 L) Drum	1

W in the catalog number indicates Winter Grade version.



Polywater J can also be applied by hand.

SPECIFICATIONS AND APPLICATIONS

- Cable Compatibility:** Passes IEEE 1210 physical and electrical testing; *Generic Requirements for Optical Cable Innerduct, Associated Conduit, and Accessories.*
- Friction Testing:** Polywater J shows superior friction reduction on a variety of jacket types.
- Physical Properties:** Contains no wax, grease, or silicone. Has no flash point and dried residue will not support combustion. Will not cement cable into conduits.
- Applications:** Long installations with multiple bends where superior friction reduction is critical; underground utilities, data centers, wastewater treatment plants and mass transit systems.*
- Temperature Use Range:** Regular Polywater J: 20°F to 120°F (-5°C to 50°C). Winter Grade Polywater WJ: -20°F to 120°F (-30°C to 50°C).

* Polywater® J is not recommended for LSZH/LSHF and certain CSPE and CPE cable jackets. For these jacket types, Polywater LZ is the recommended/approved product.

polywater.com | +1 651-430-2270

IMPORTANT NOTICE: The statements here are made in good faith based on tests and observations we believe to be reliable. However, the completeness and accuracy of the information is not guaranteed. Before using, the end-user should conduct whatever evaluations are necessary to determine that the product is suitable for the intended use.

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Polywater®
Solutions at work.

American Polywater Corporation
11222 60th Street N.
Stillwater, MN 55082 USA

+1 651-430-2270 Main
+31 10 233 0578 Europe, Middle East, N. Africa
email: support@polywater.com

DESCRIPTION

Polywater® LZ Lubricant is a high-performance, cable pulling lubricant. Polywater LZ is compatible with a broad variety of LSZH/LSHF compounds. Polywater LZ is also compatible with other high-performance cable jackets. It provides excellent tension reduction and is recommended for all types of cable pulling.

Polywater LZ is slow drying and leaves a thin, slippery film that retains its lubricity for months after use. Polywater LZ does not sustain flame when used with fire-retardant cables and systems. Its dried residue is nonconductive and noncombustible.

Polywater LZ is a stringy gel. It can be applied by hand or using Polywater's LP-D5 Pump. It is also available in the unique Front End Pack™ prelubrication bags.

FRICTION TESTING

Lubricity: Polywater LZ shows superior friction reduction on a variety of jacket types. Typical friction coefficients at 200 lbs/ft (2.91 kN/m) normal pressure are shown. Test results are based on the method described in the white paper, "[Coefficient of Friction Measurement on Polywater's Friction Table, 2019](#)". Values are averages based on cable jacket and conduit materials from multiple manufacturers.

CABLE JACKET	CONDUIT TYPE				
	STEEL	FRP	HDPE	PVC	EMT
LSZH	.16	.17	.07	.08	.21
CSPE	.21	.24	.12	.16	.24
CPE	.15	.19	.09	.10	.17
XLPE	.13	.12	.06	.06	.12
LLDPE	.10	.11	.05	.06	.13

Coefficient of friction data on additional or specific cable jackets or conduits can be obtained from American Polywater Corporation.



Polywater LZ is a specification grade lubricant

PRODUCT FEATURES

- **Low Friction Coefficient:** Maximum tension reduction on all types of cable jackets.
- **Universal:** Suitable for all types of jackets and cable, including power, control, and instrumentation cable.
- **Low-Smoke Zero Halogen (LSZH/LSHF) Compatible:** Extensively tested on LSZH/LSHF thermoplastic and thermoset jackets.

END USE

Polywater LZ is a specification grade lubricant that meets the performance requirements of:

- Nuclear and other generation plants
- Mass transit systems and airports
- Oil and petrochemical

OFFICIAL APPROVALS

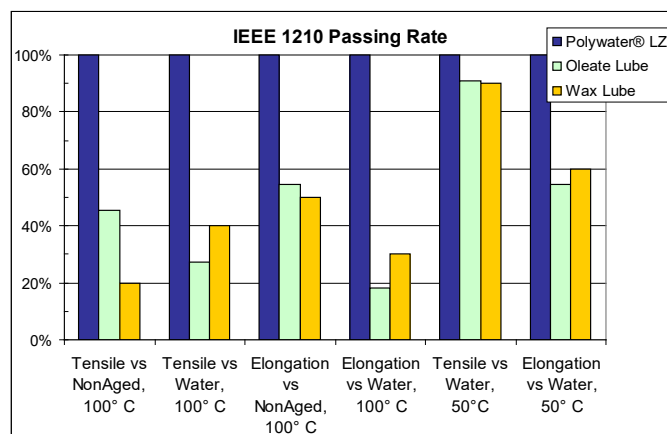
UL Listed
UL Listed to Canadian safety standards

CABLE COMPATIBILITY

Tensile and Elongation:

LSZH, CSPE, LLDPE, XLPE, CPE, and PVC cable jacket materials aged in Polywater Lubricant LZ per IEEE Standard 1210¹ meet the tensile and elongation retention requirements of that standard.

Modern LSZH jackets are numerous and vary significantly in formulation. Polywater LZ shows broad compatibility with this jacket technology. As shown in the graph below, the common cable pulling lubricants available through local supply houses show significant and sometimes devastating effects on LSZH cable jackets.



Polyethylene Stress Cracking:

Polywater LZ shows no stress cracking on LDPE, MDPE, or HDPE cable jackets when tested per IEEE Standard 1210.¹

Volume Resistivity:

There are no significant changes in the conductive properties of XLPE and EPR semi-conducting compounds when volume resistivity is tested according to IEEE Standard 1210.¹

Building Wire Testing:

THHN and XLPE building wire meet UL tensile, elongation, and voltage withstand requirements after exposure to Polywater LZ as tested by UL requirements.²

Cable Approvals:

Polywater LZ is approved by many cable manufacturers. Contact American Polywater for details.

¹ IEEE Std 1210-2004, IEEE Standard Tests for Determining Compatibility of Cable-Pulling Lubricants with Wire and Cable.

² UL Subject 267, Investigation for Wire-Pulling Compounds.

PHYSICAL PROPERTIES

PROPERTY	RESULT
Appearance	White, stringy gel
% Nonvolatile solids (weight)	4.0
VOC content	0 gms/L 200 gms/L (Winter Grade)
Viscosity (Brookfield)	35,000-50,000 cps @10 rpm
pH	6.5-7.5

PERFORMANCE PROPERTIES

Cling factor is a measure of the ability to apply the lubricant and have it stay on the jacket while the cable enters the conduit.

A six-inch length (152 mm) of a one-inch (25 mm) diameter cable will hold at least 35 grams of Polywater LZ for one minute when held vertically at 70°F (21°C).

Coatability:

Coatability is a measure of the lubricant's ability to coat the jacket as a thin film for continued lubricity on longer pulls.

Polywater LZ will wet out evenly on cable jacket surfaces. It will not bead up or rub off the jacket sample. A one-inch (25 mm) diameter XLPE cable dipped six inches (152 mm) into Polywater LZ, then withdrawn and held vertically, will retain at least 25 grams of Polywater LZ for one minute at 70°F (21°C).

Combustibility:

Combustibility is a measure of combustion properties of the lubricant residue in a fire situation (with an impinging heat flux).

Polywater LZ has no flash point and its dried residue will not support combustion and spread flame. A 15-gram sample of the Polywater LZ, when placed in a one-foot, split, metal conduit and fully dried for 24 hours at 105°C, will not ignite and spread a flame more than three inches beyond the point of ignition when subjected to a continuous heat flux of 85 kW/m². The total test time was 30 minutes.

Test method described in "Fire Parameters and Combustion Properties of Cable Pulling Compound Residues," presented to the International Wire & Cable Symposium, 1987.

APPLICATION PROPERTIES

Application Systems:

Polywater LZ has a stringy gel consistency that makes it easy to lift, carry, and hand-apply.

Polywater LZ can also be pumped directly into the conduit or onto the cable using the Polywater LP-D5 specialty lubricant pump. This allows hands-free transfer and consistent application of lubricant. Polywater's low-shear pump will not change the gel character of Polywater LZ. The LP-D5 pump applies lubricant at a rate of 1-2 gallons (4-8 liters) per minute.

Polywater LZ Front End Packs are bag packages that "prelubricate" the head end of the cable during the pull. The Front End Pack attaches to the winch line and prelubricates as it goes through the conduit. Two sizes are available to fit 2" and larger conduits.

Pull-Planner™ Tension Calculation Software is available from Polywater. Pulling tension estimations can ensure the use of appropriate pulling equipment and that the cable is installed within safe limits.

Polywater LZ is also available in a special-order, pourable version (lower viscosity) called Polywater PLZ.

Temperature Use Range:

Polywater LZ:

20°F to 120°F (-5°C to 50°C).

Polywater WLZ (Winter Grade version):

-20°F to 120°F (-30°C to 50°C)

Temperature Stability:

Polywater LZ will not phase-out or separate after five freeze/thaw cycles or 5-day exposure at 120°F (50°C).

Clean-up:

Polywater LZ is nonstaining. Complete clean-up is possible with water.

Storage and Shelf Life:

Store Polywater LZ in a tightly sealed container away from direct sunlight. Shelf life is 24 months.

ENVIRONMENTAL PROPERTIES

Aquatic Toxicity:

Polywater LZ is safe in the aquatic environment and passes CCR Title 22 Fathead Minnow Hazardous Waste Screen Bioassay
Result: PASS (LC50 > 750 mg/L)

DIRECTIONS FOR USE

Polywater LZ can be hand-applied or pumped onto the cable as it enters the conduit.

For long pulls, place approximately two-thirds of the recommended quantity of lubricant into the conduit using the Front End Packs or by pumping.

For Front End Packs use, attach the packs of Polywater LZ to the winch line or pulling rope in front of the cable by using tape or cable ties. Start the pull and slit open the entire length of the pack(s) with a sharp knife as it enters the conduit.

Supplement with direct jacket lubrication as the cable enters the conduit.

Clean up by wiping off any excess lubricant with a rag.

Recommended Lubricant Quantity:

$$Q = k \times L \times D$$

Where:

Q = quantity in gallons (liters)

L = length of conduit run in feet (meters)

D = ID of the conduit in inches (mm)

k = 0.0015 (0.0008 if metric units)

The quantity that is appropriate for any given pull can vary from this recommendation by 50%, depending on the complexity of the pull. Consider the following factors:

Cable weight and stiffness
(Increase quantity for stiff, heavy cable)

Conduit condition
(Increase quantity for old, dirty, or rough conduits)

Conduit fill
(Increase quantity for high percent conduit fill)

Number of bends
(Increase quantity for pulls with several bends)

Pulling environment
(Increase quantity for high temperatures)

MODEL SPECIFICATION

The statement below may be inserted into a specific job specification to help maintain engineering standards and ensure project integrity.

The cable pulling lubricant shall be Polywater® LZ Lubricant. The lubricant shall be UL (or CSA) listed. The cable pulling lubricant shall provide excellent friction reduction with good cling and wetting through long pulls and multiple bends. The lubricant shall leave minimal, noncombustible residue. It shall be compatible with most cable jacket materials and be extensively tested on a broad variety of low smoke, halogen-free cable jacket materials.

Cable jacket compatibility shall be tested with the specific LSZH jacket material used on the cable. Test data shall be provided by the cable manufacturer or the lubricant manufacturer. It shall not stress-crack polyethylene per ASTM Standard 1693. There shall be no significant changes in the conductive properties of XLPE and EPR semi-conducting compounds when the lubricant’s effect on volume resistivity is tested according to IEEE Standard 1210.

A 15-gram sample of the lubricant, when placed in a one-foot, split metal conduit and fully dried for 24 hours at 105 degrees C, shall not spread a flame more than three inches beyond a point of ignition at a continued heat flux of 85 kW/meter². Total time of test shall be 30 minutes.

ORDER INFORMATION

CAT #	PACKAGE DESCRIPTION
LZ-55	½-gal. bag in a box (1.9 liters) 6/case
LZ-110	½-gal. bag in a pail (1.9 liters) 10/pail
LZ-35	1-qt. squeeze bottle (.95 liter) 12/case
LZ-128	1-gal. pail (3.78 liter) 4/case
LZ-640	5-gal. pail (18.9 liter)
LZ-DRUM	55-gal. drum (208 liter)
	Winter Grade
WLZ-55	½-gal. bag in a box (1.9 liters) 6/case
WLZ-110	½-gal. bag in a pail (1.9 liters) 10/pail
WLZ-35	1-qt. squeeze bottle (.95 liter) 12/case
WLZ-128	1-gal. pail (3.78 liter) 4/case
WLZ-640	5-gal. pail (18.9 liter)

CONTACT US

+1-651-430-2270 Main | +31 10 233 0578 Europe & Africa | +971 4 5521709 APAC & GCC | email: support@polywater.com

IMPORTANT NOTICE: The statements here are made in good faith based on tests and observations we believe to be reliable. However, the completeness and accuracy of the information is not guaranteed. Before using, the end- user should conduct whatever evaluations are necessary to determine that the product is suitable for the intended use. American Polywater expressly disclaims any implied warranties and conditions of merchantability and fitness for a particular purpose. American Polywater’s only obligation shall be to replace such quantity of the product proven to be defective. Except for the replacement remedy, American Polywater shall not be liable for any loss, injury, or direct, indirect, or consequential damages resulting from product’s use, regardless of the legal theory asserted.

